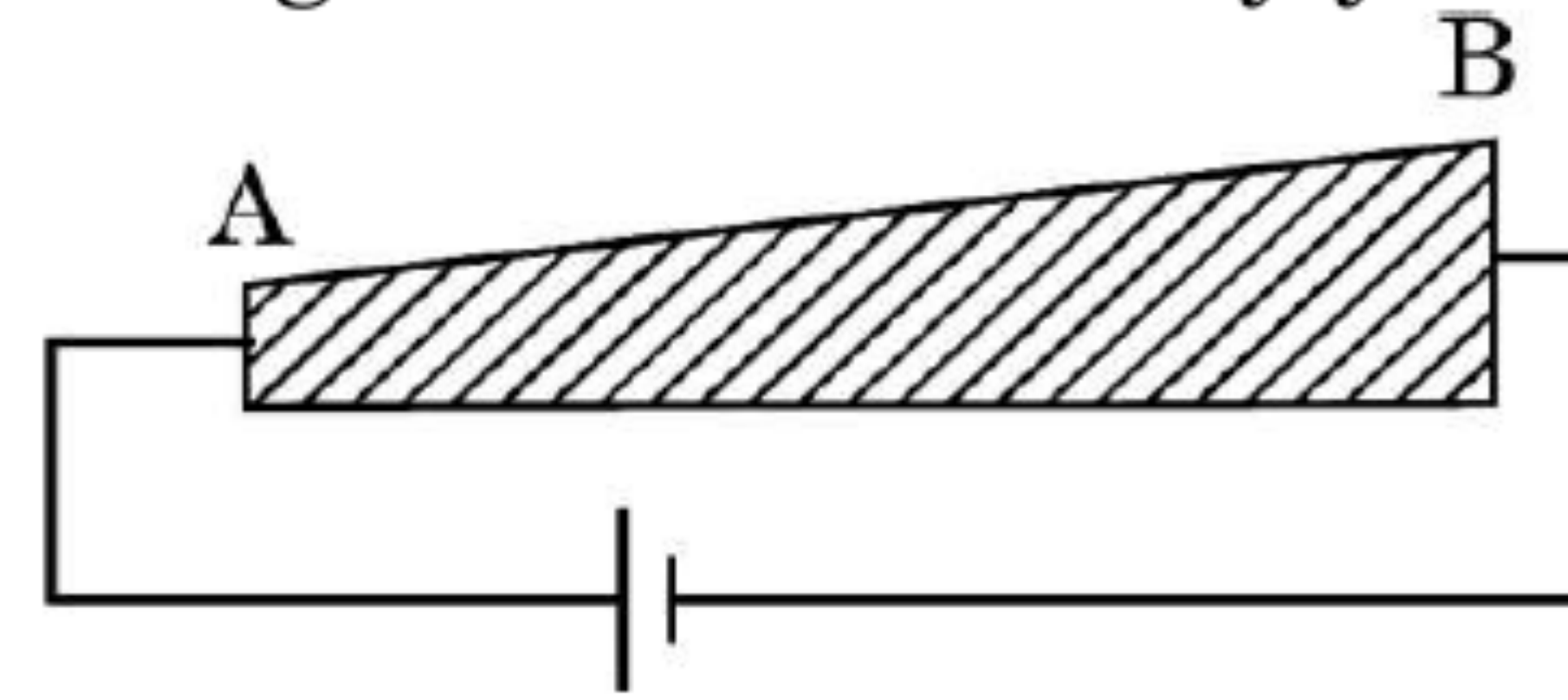
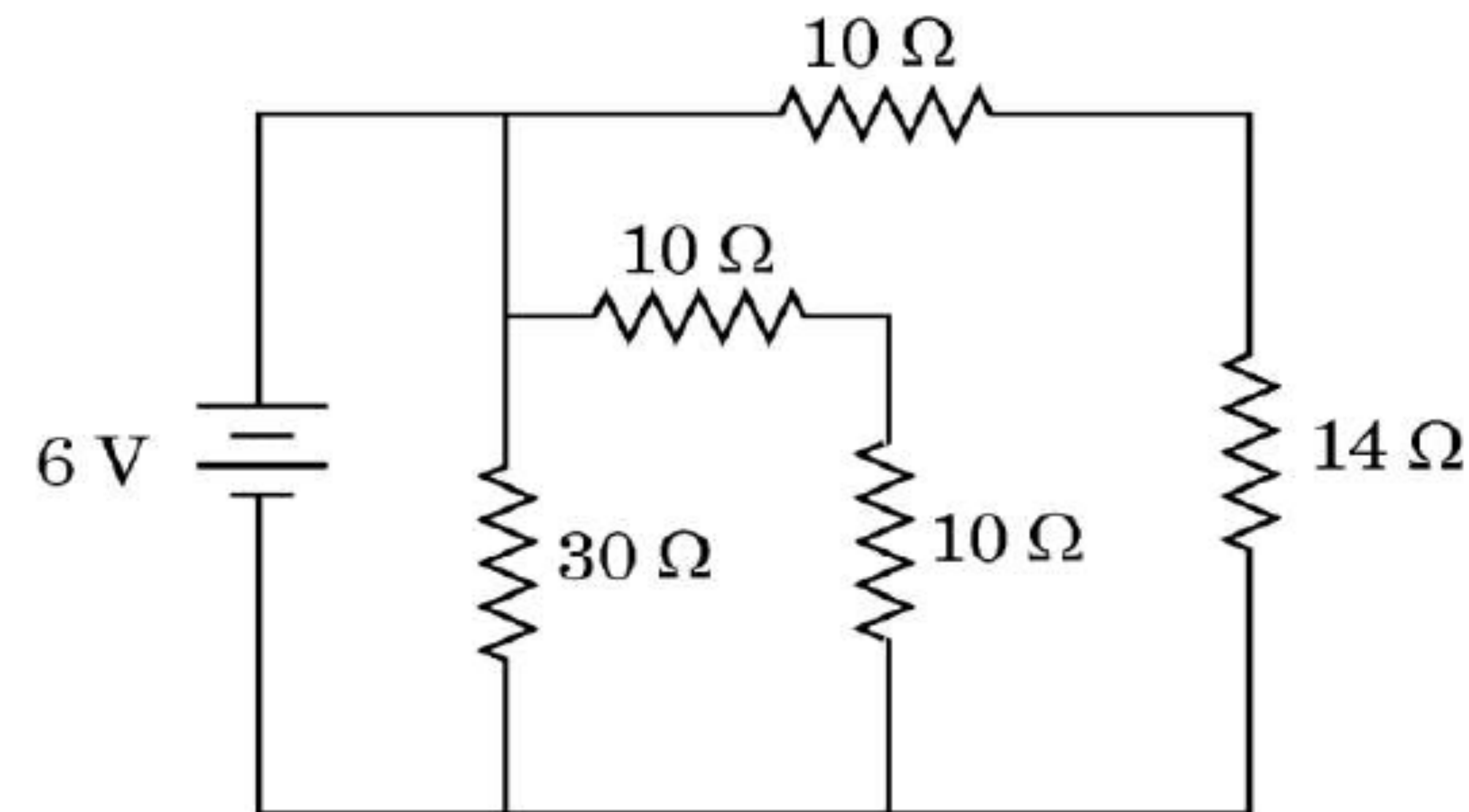


32. (a) (i) Define mobility of electrons. Give its SI units.

(ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire ? Justify your answer.



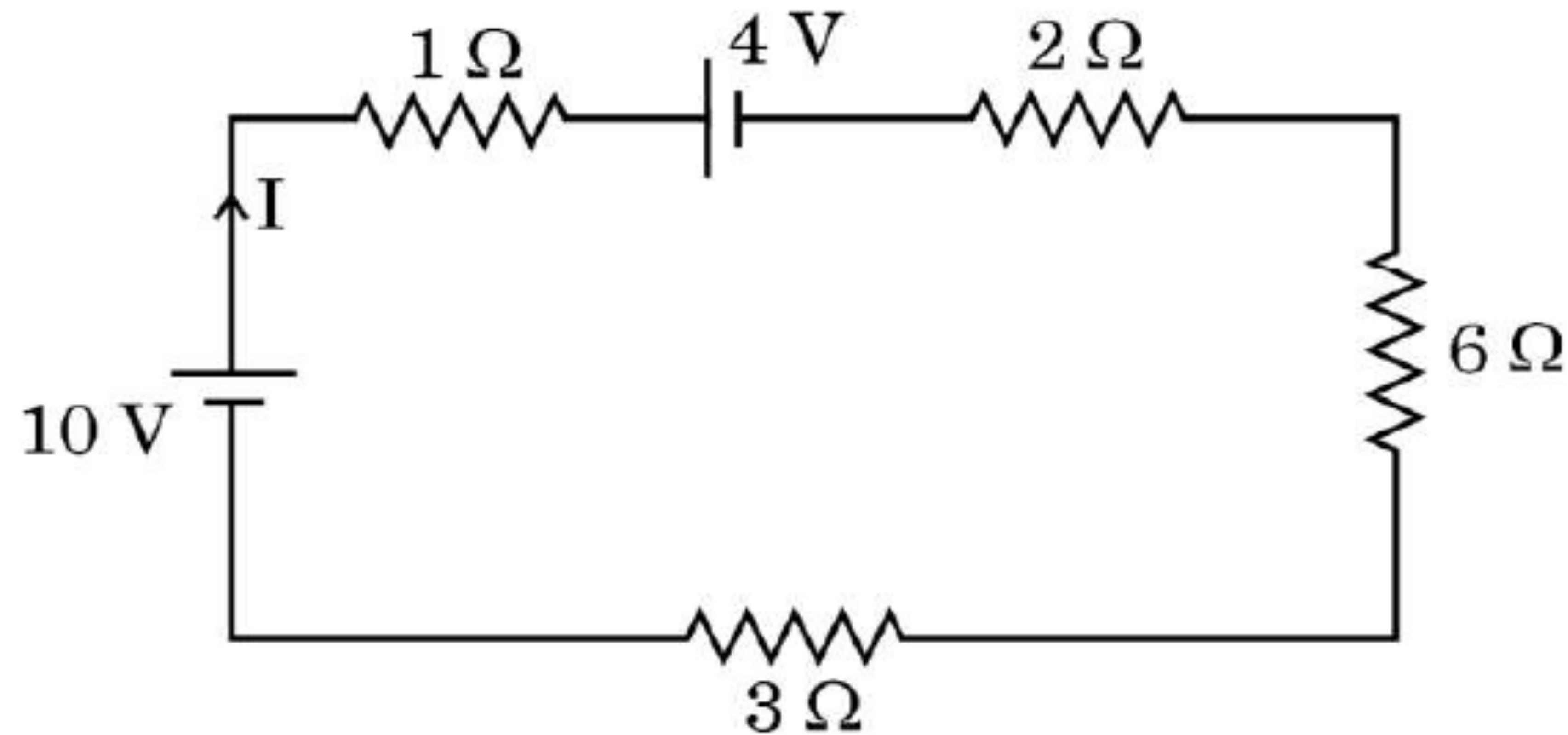
(iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.



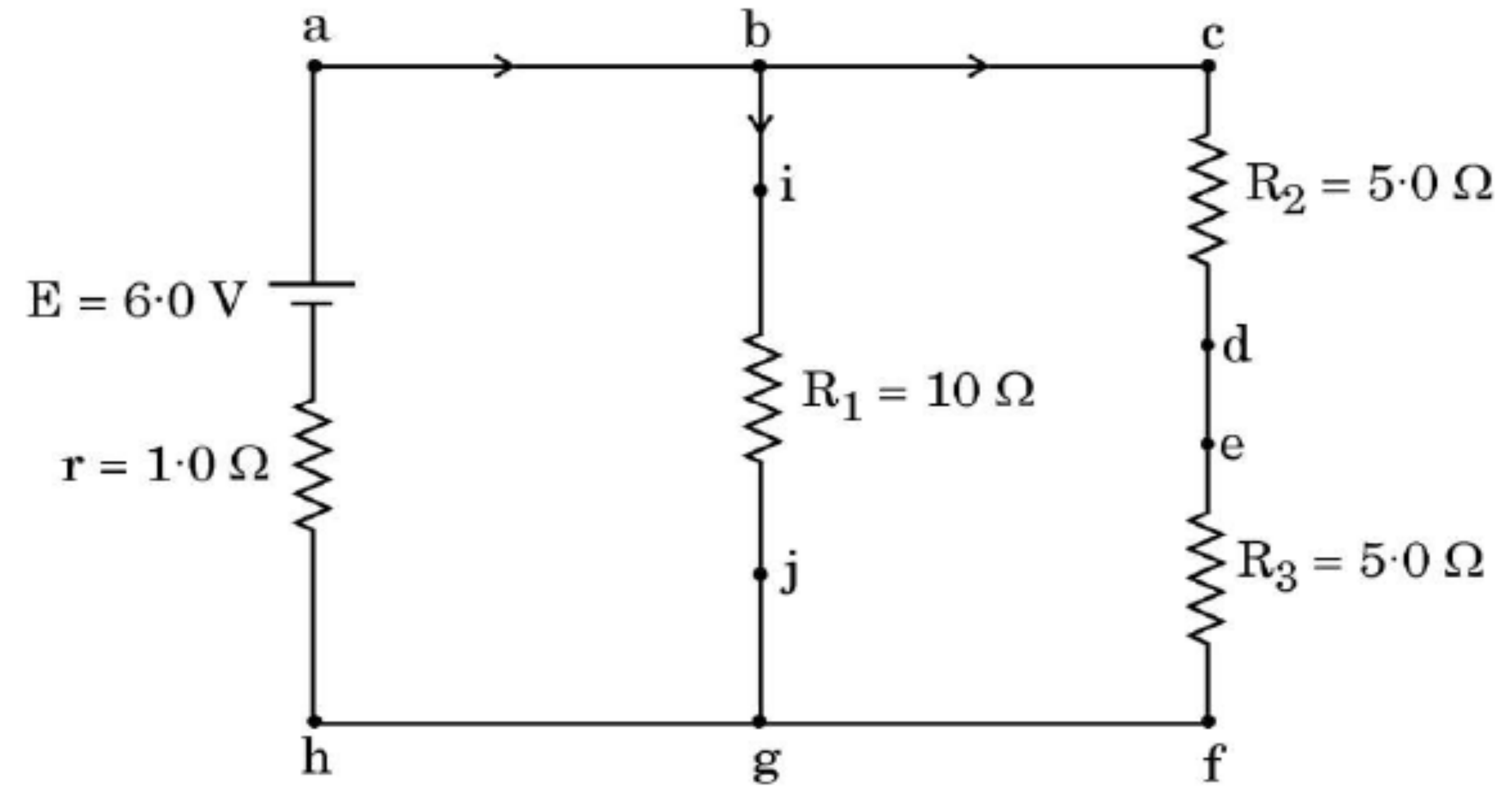
OR

- (b) (i) Define electrical conductivity of a wire. Give its SI unit.
- (ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries ?
- (iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.

5



34. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



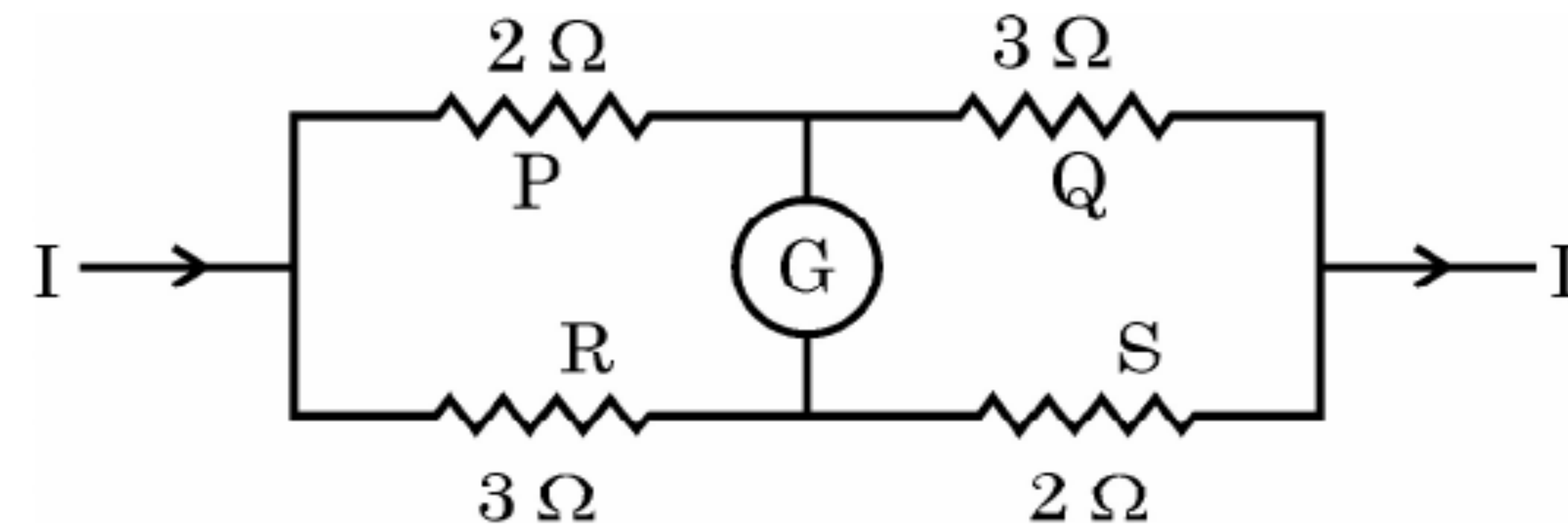
Answer the following questions based on the above :

- | | |
|---|---|
| (a) Which points are at the same potential in the circuit ? | 1 |
| (b) What is the current through arm bg ? | 1 |
| (c) Find the potential difference across resistance R_3 . | 2 |

OR

- | | |
|--|---|
| (c) What is the power dissipated in resistance R_2 ? | 2 |
|--|---|

17. **Assertion (A)** : The given figure does not show a balanced Wheatstone bridge.

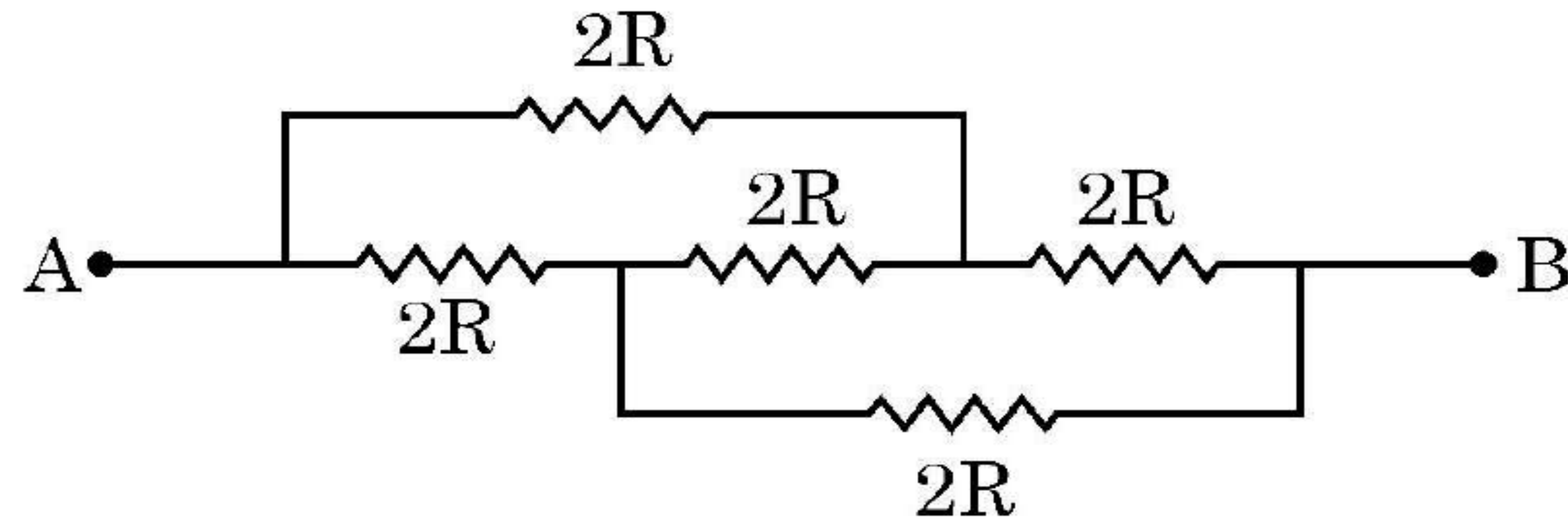


Reason (R) : For a balanced bridge small current should flow through the galvanometer.

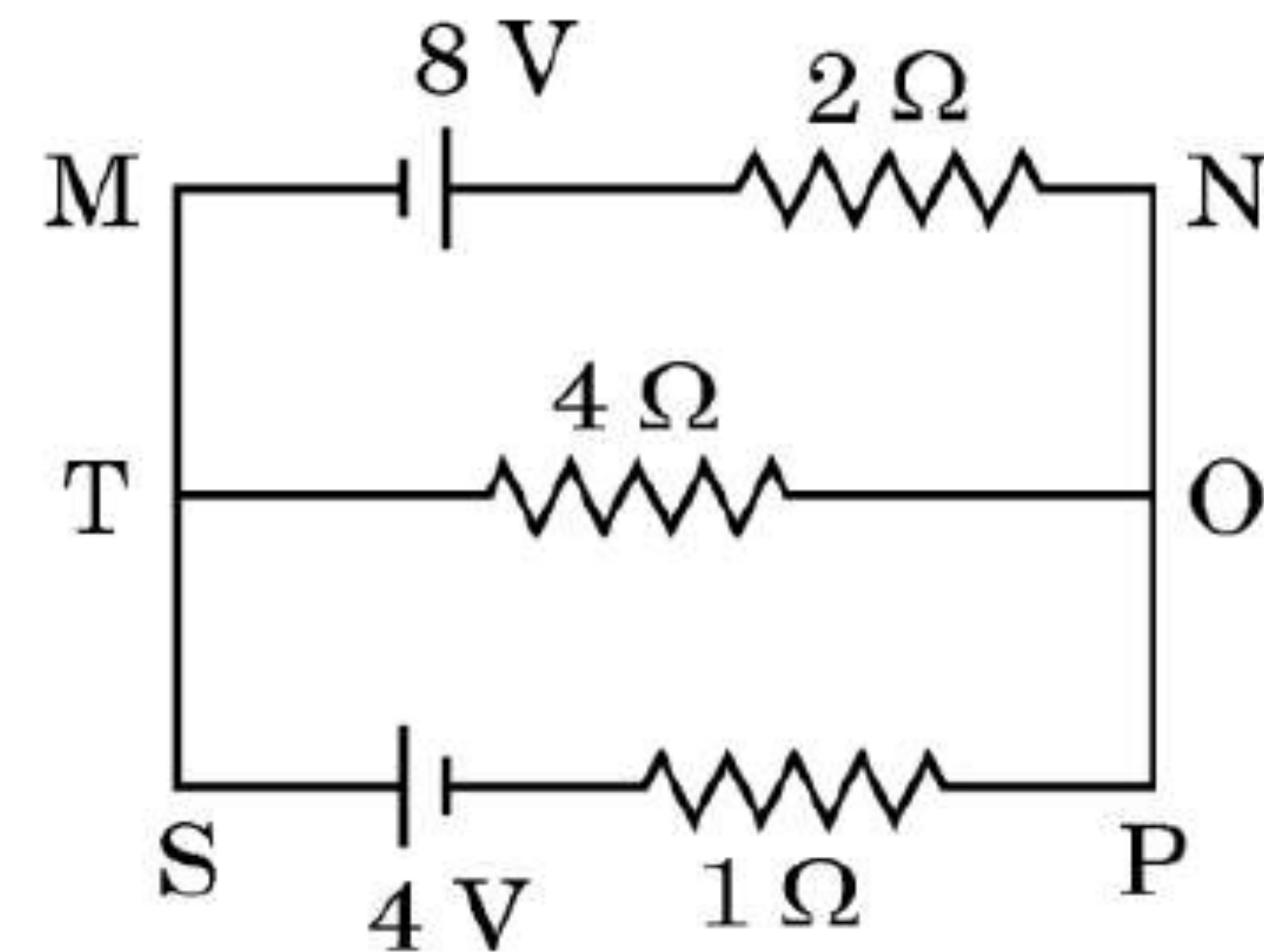
17. **Assertion (A)** : The equivalent resistance between points A and B in the given network is $2R$.

1

Reason (R) : All the resistors are connected in parallel

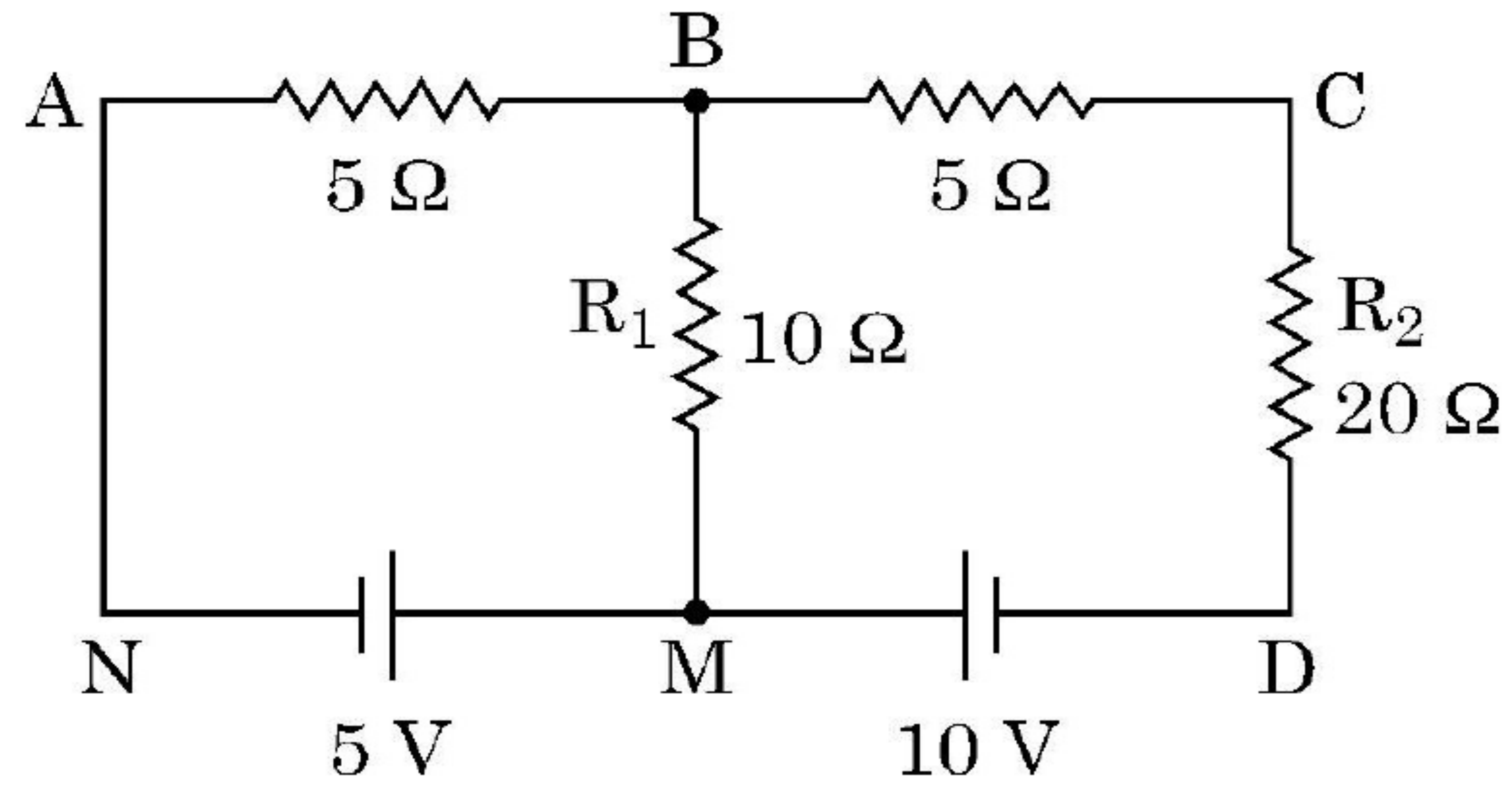


- (b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.
- (ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure.



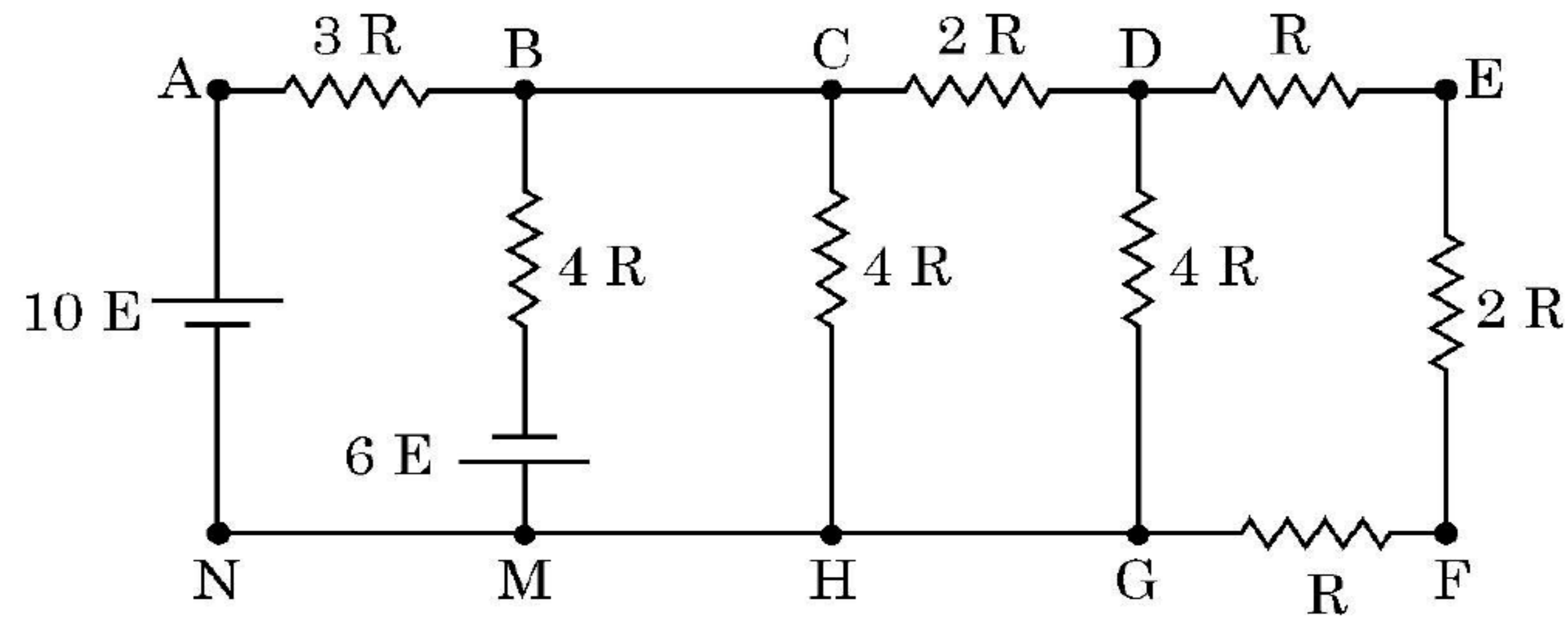
27. Find the currents flowing through the branches AB and BC in the network shown.

3



25. Find the current in branch BM in the network shown :

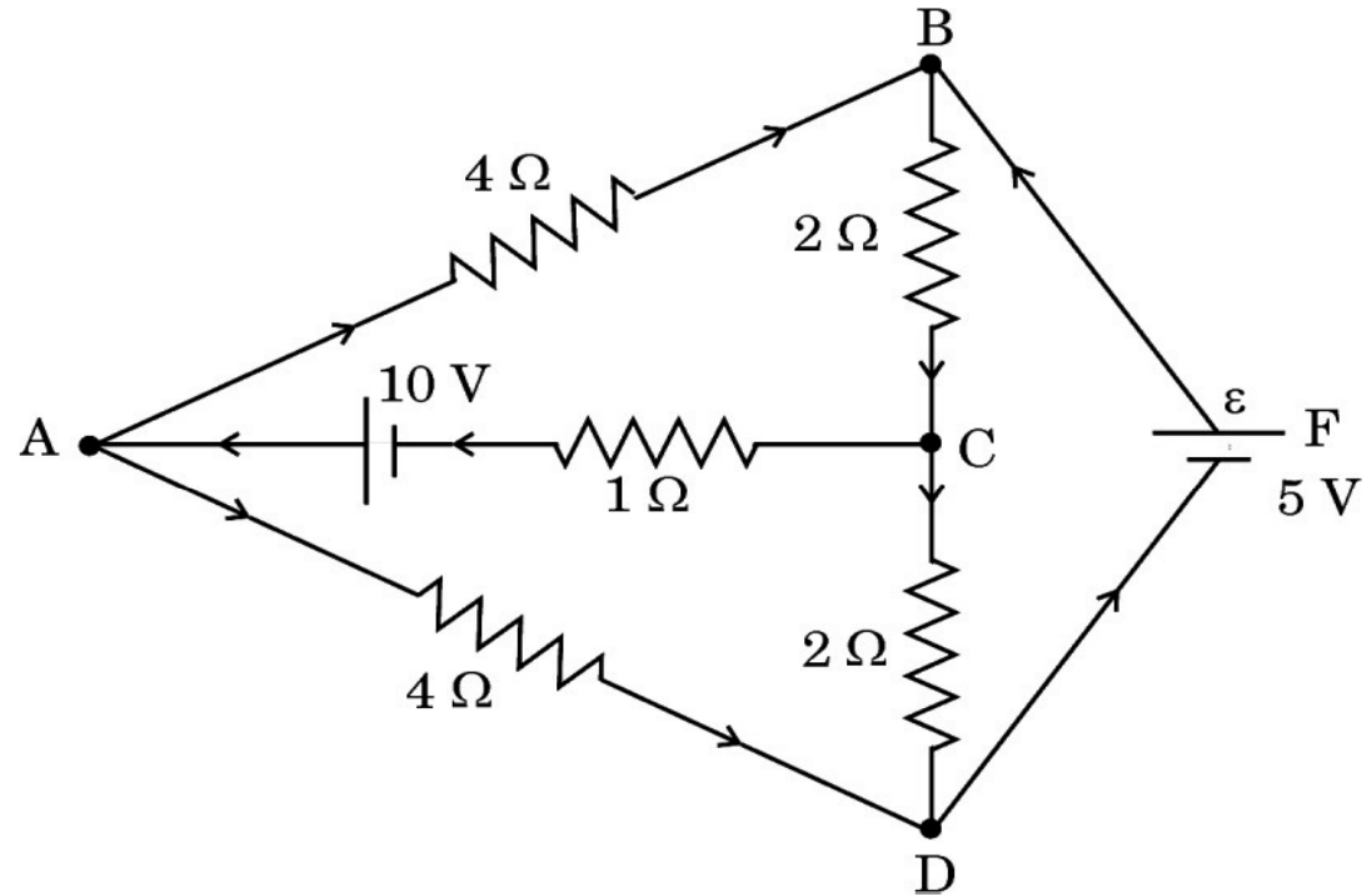
3



17. (a) What is meant by 'relaxation time' of free electrons in a conductor ?
Show that the resistance of a conductor can be expressed by
$$R = \frac{m l}{n e^2 \tau A}$$
, where symbols have their usual meanings. 2
- OR**
- (b) Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it. 2

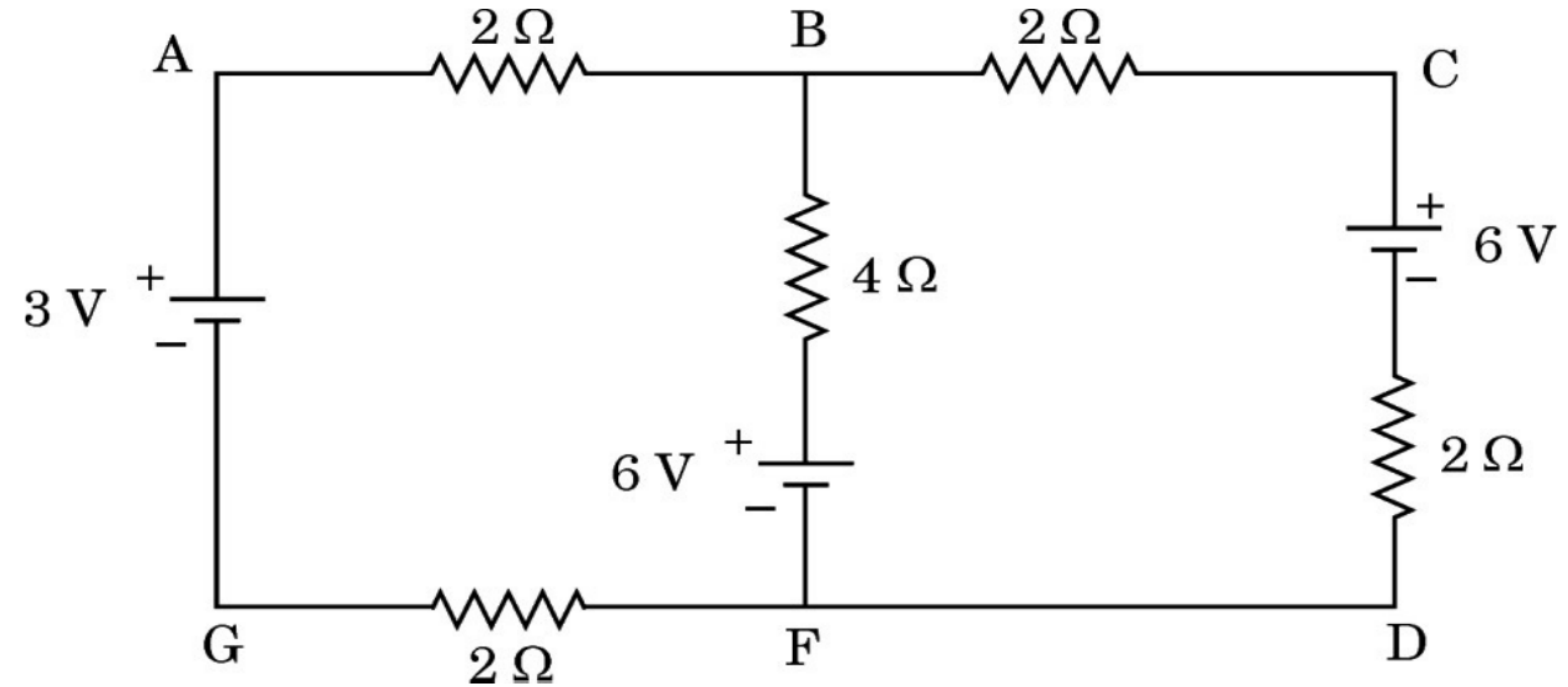
22. Determine the current in branches AB, AC and BC of the network shown in figure.

3



- 25.** The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD.

3

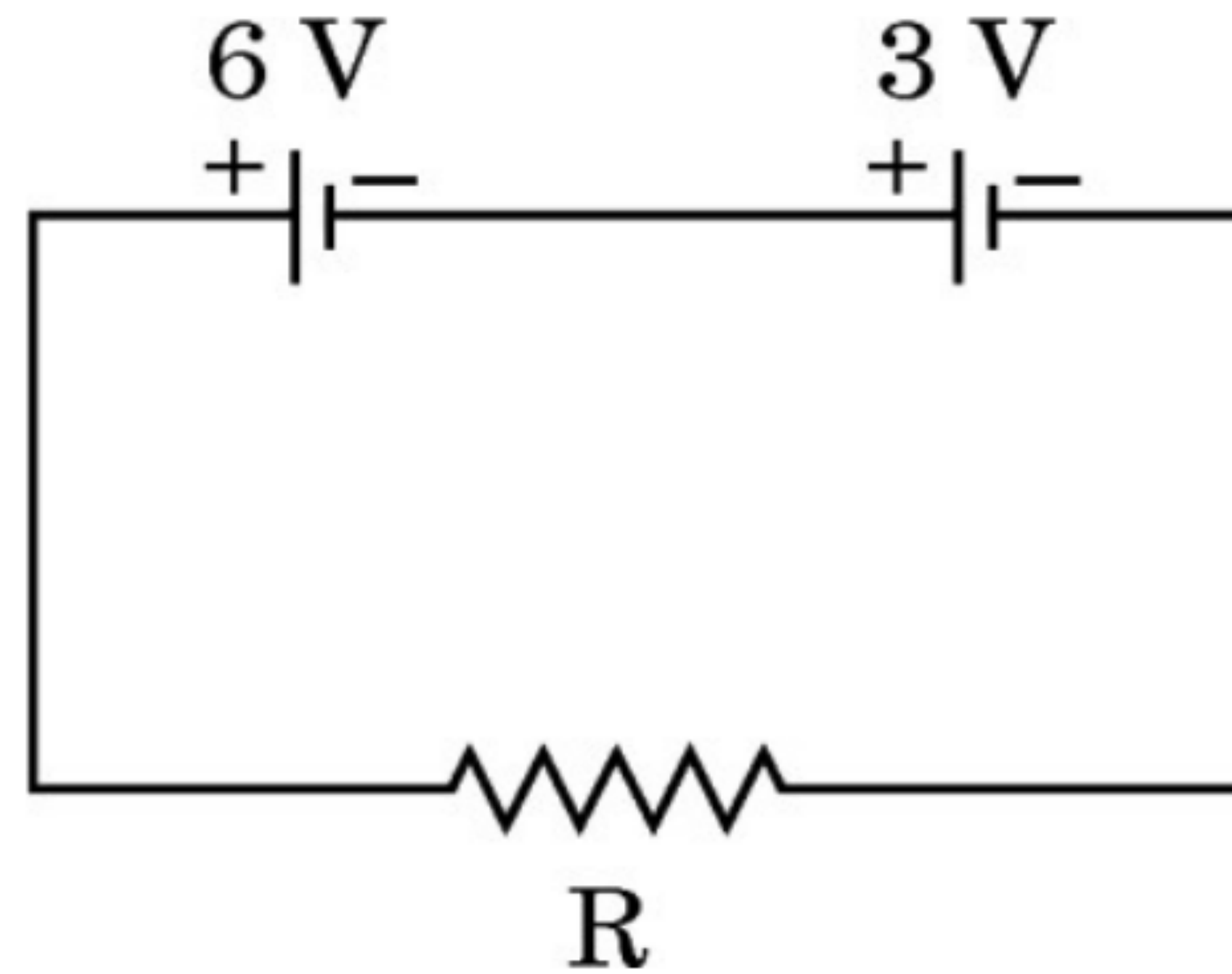


- 23.** (a) Define current density. Is it a scalar or a vector ? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m , charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\frac{ne^2}{m} \right) \tau$. 3

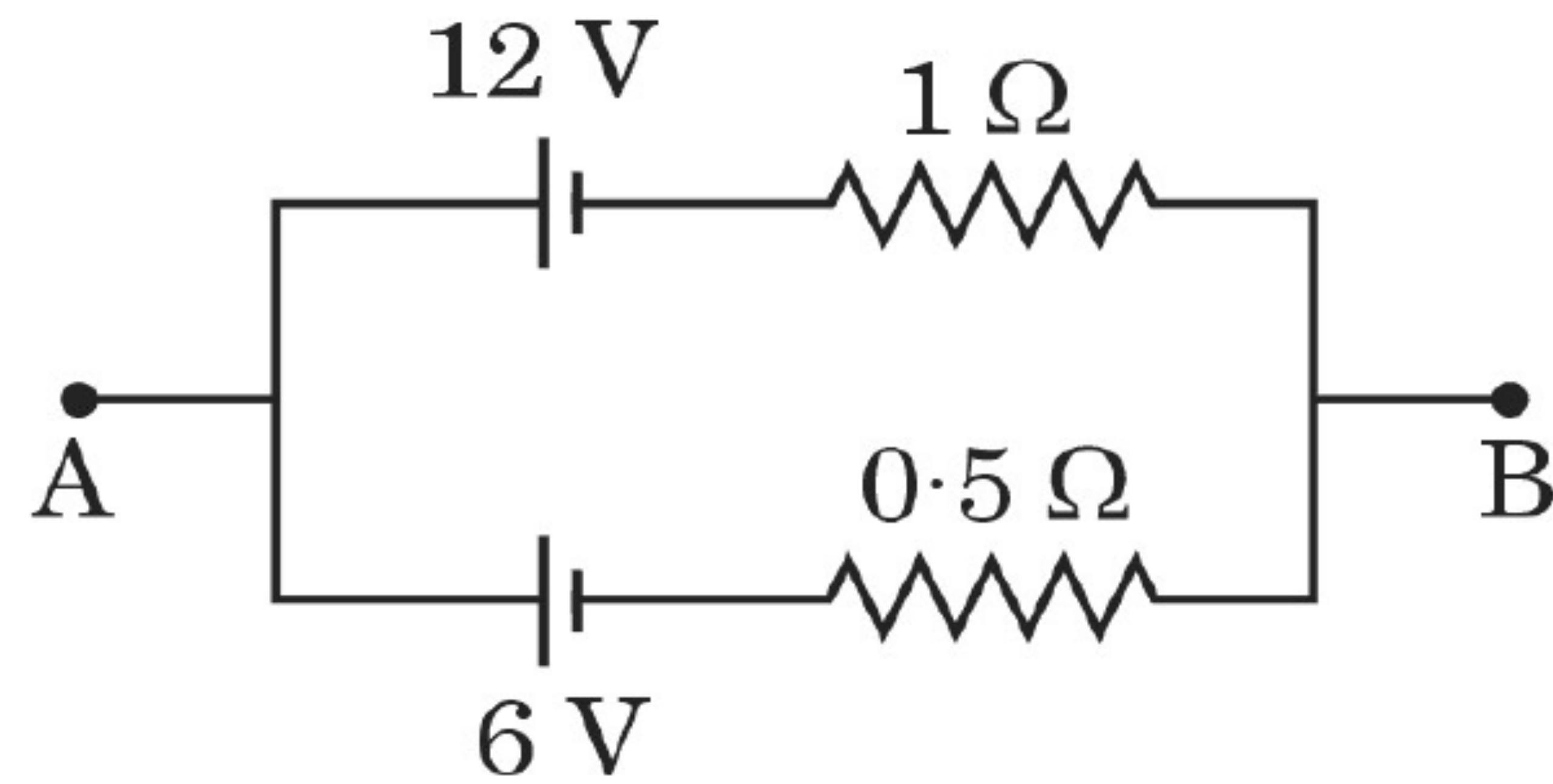
OR

- (b) What is a Wheatstone bridge ? Obtain the necessary conditions under which the Wheatstone bridge is balanced. 3

17. Two batteries of emfs 6 V and 3 V and internal resistances $0.8\ \Omega$ and $0.2\ \Omega$ respectively are connected in series to an external resistance R , as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero.



3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

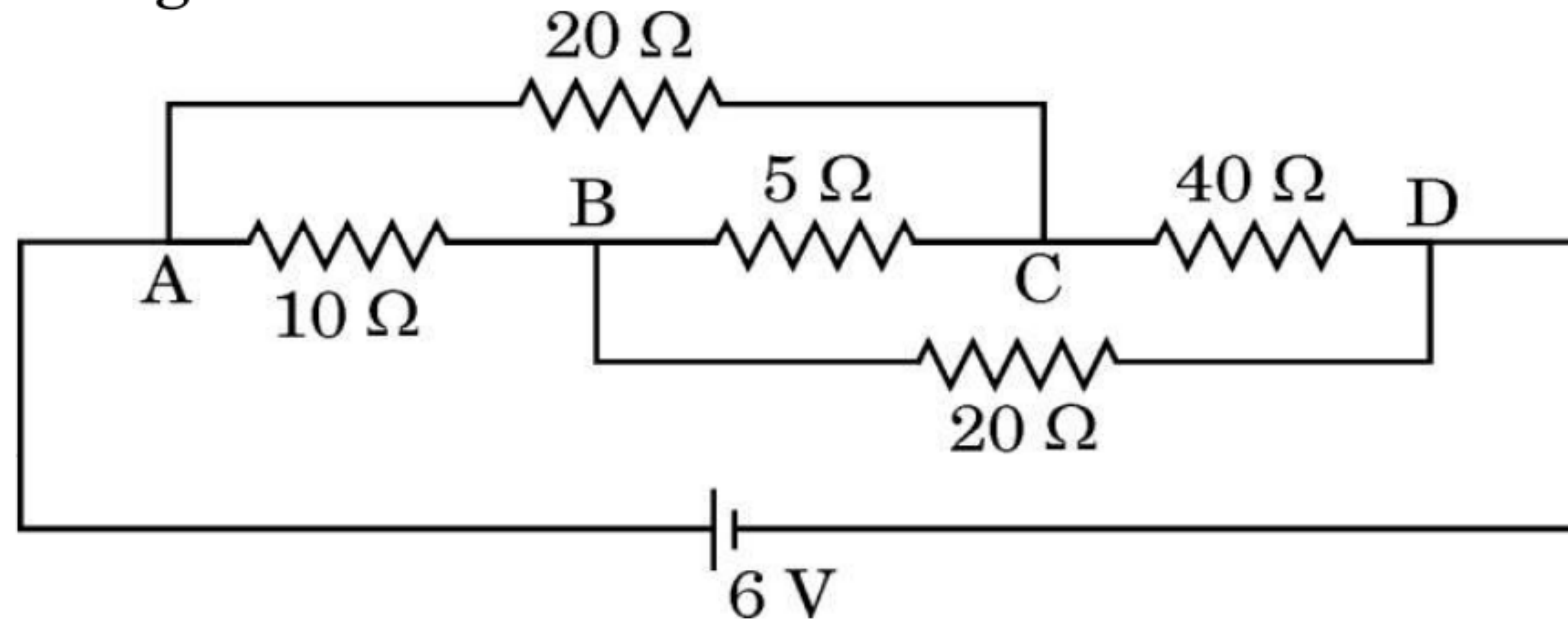
(B) 8 V

(C) 9 V

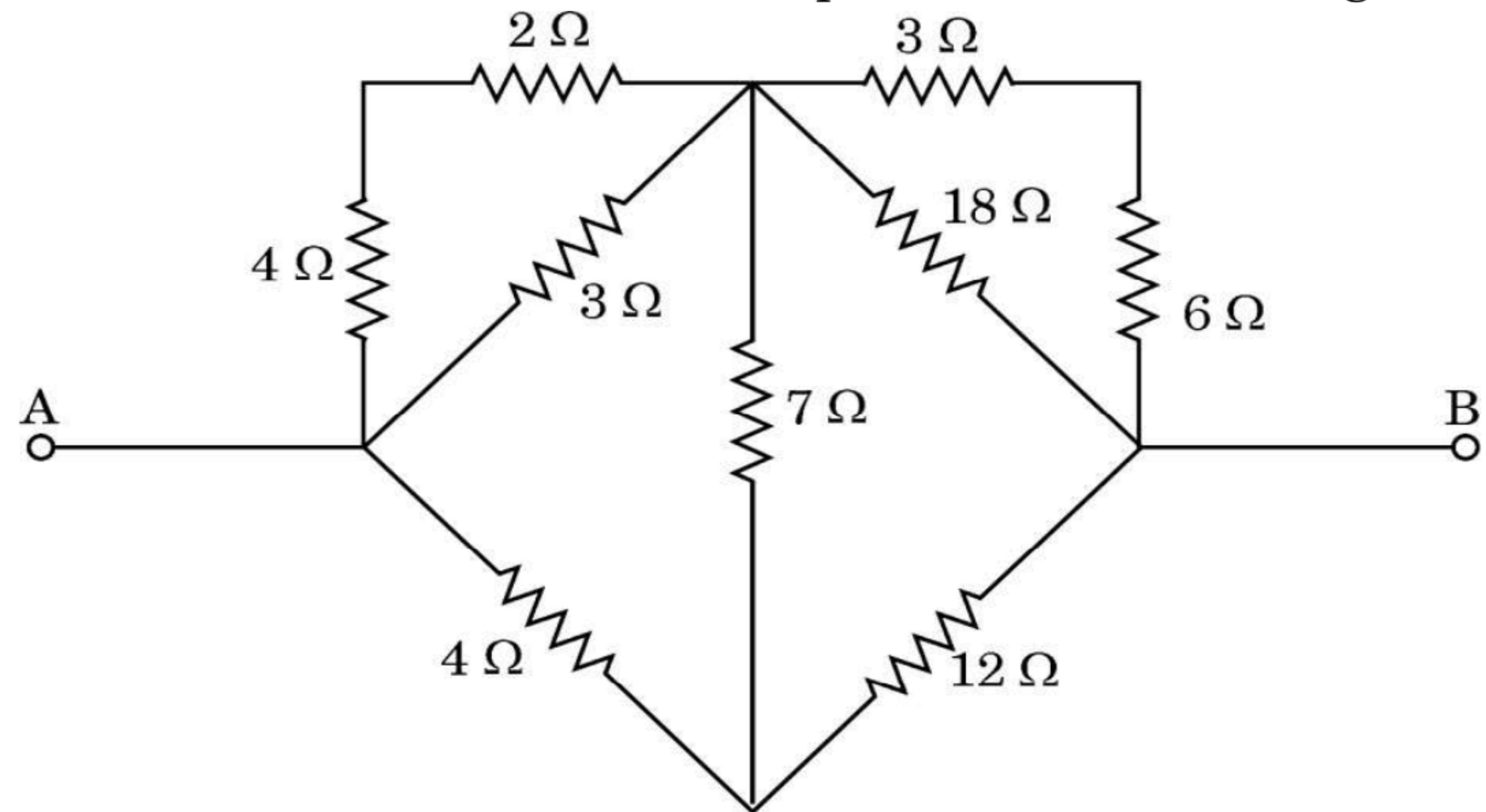
(D) 12 V

17. Calculate the value of the current passing through the battery in the given circuit diagram.

2



7. The effective resistance between points A and B in the given circuit is :



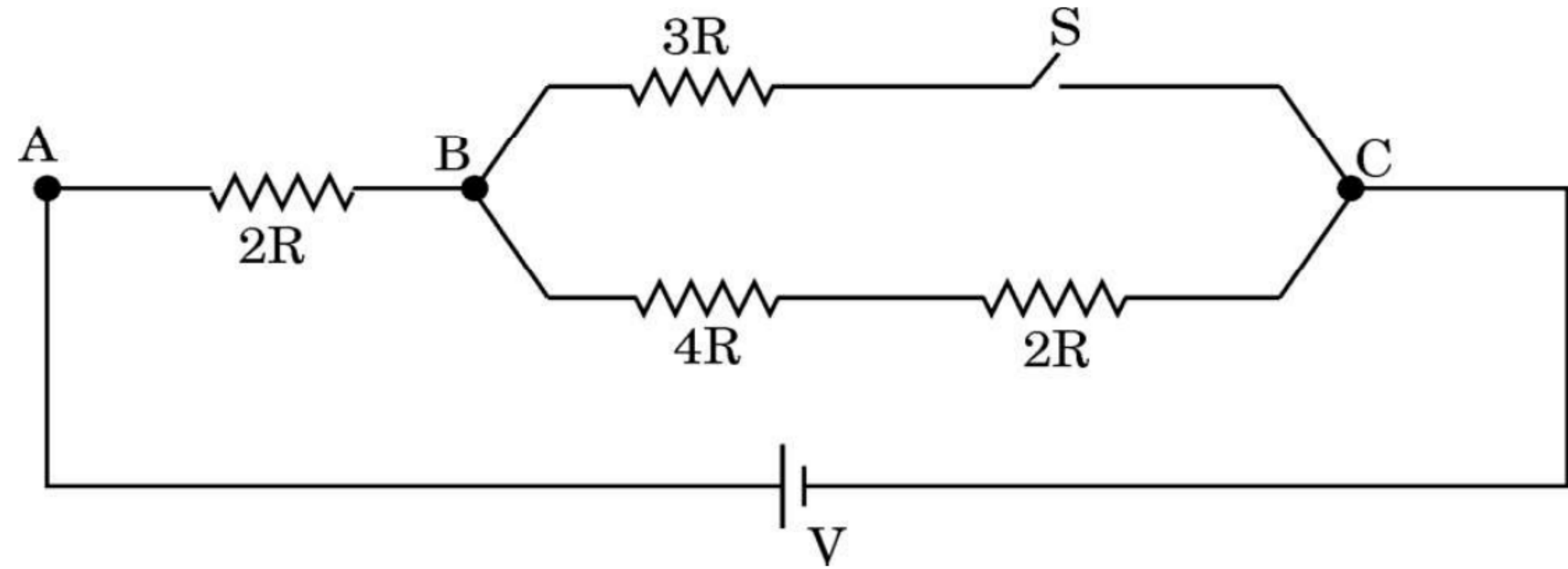
(A) $6\ \Omega$

(B) $\frac{8}{3}\ \Omega$

(C) $\frac{16}{3}\ \Omega$

(D) $2\ \Omega$

1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



(A) $\frac{1}{4}$

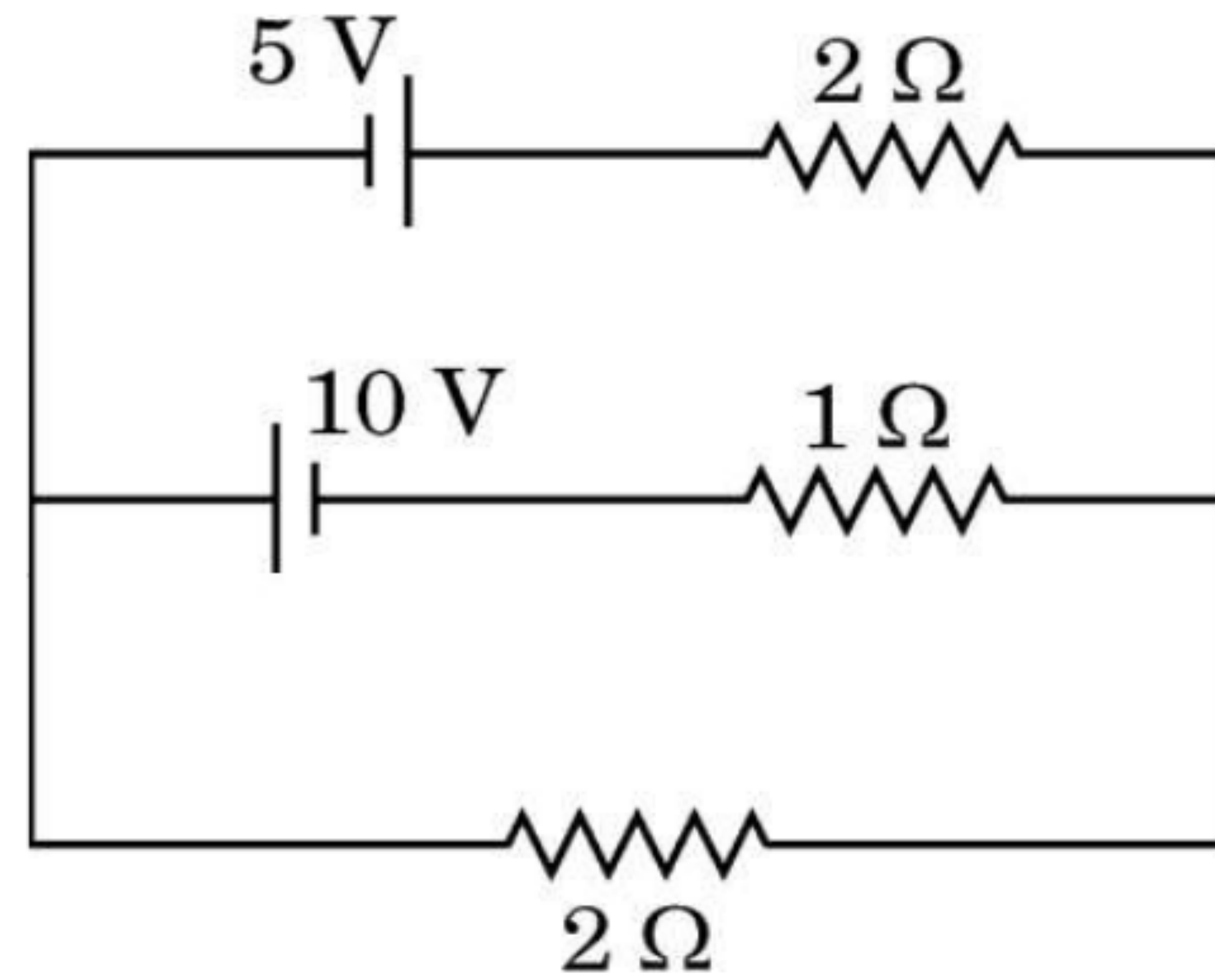
(B) $\frac{1}{2}$

(C) 1

(D) 2

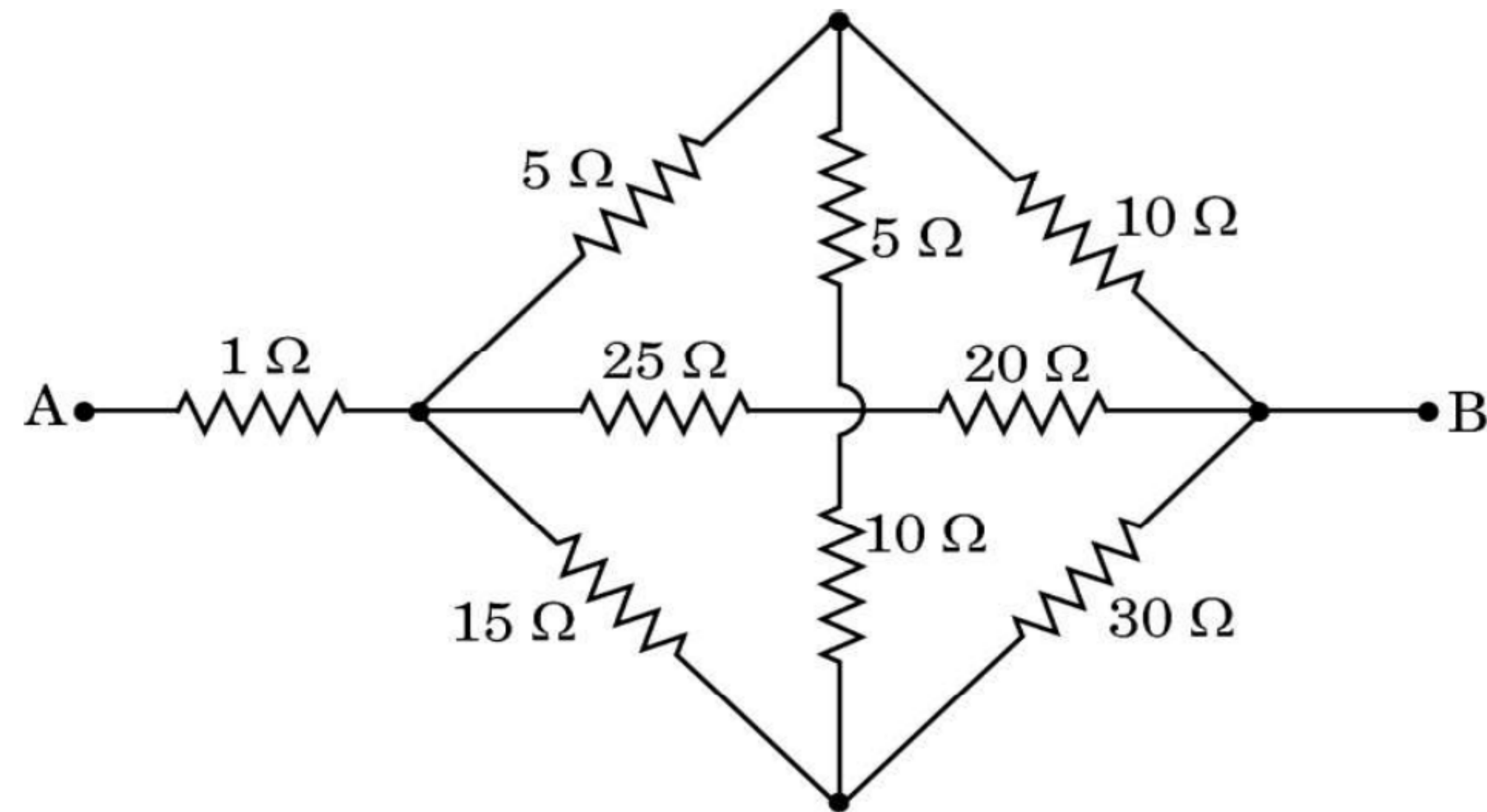
- 26.** State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit.

3

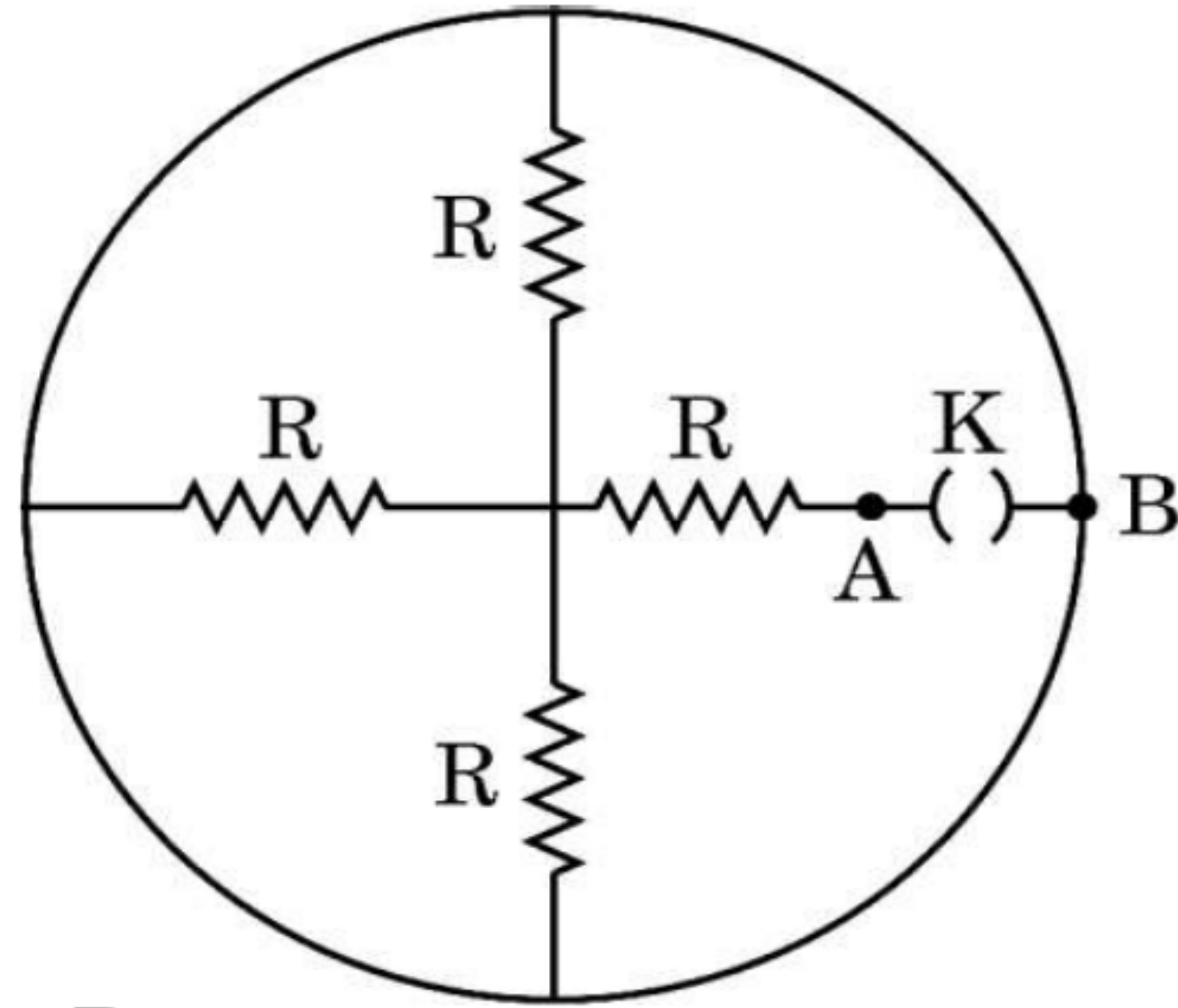


17. Find the equivalent resistance between points A and B for the network shown in the figure.

2



5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



(A) $4R$

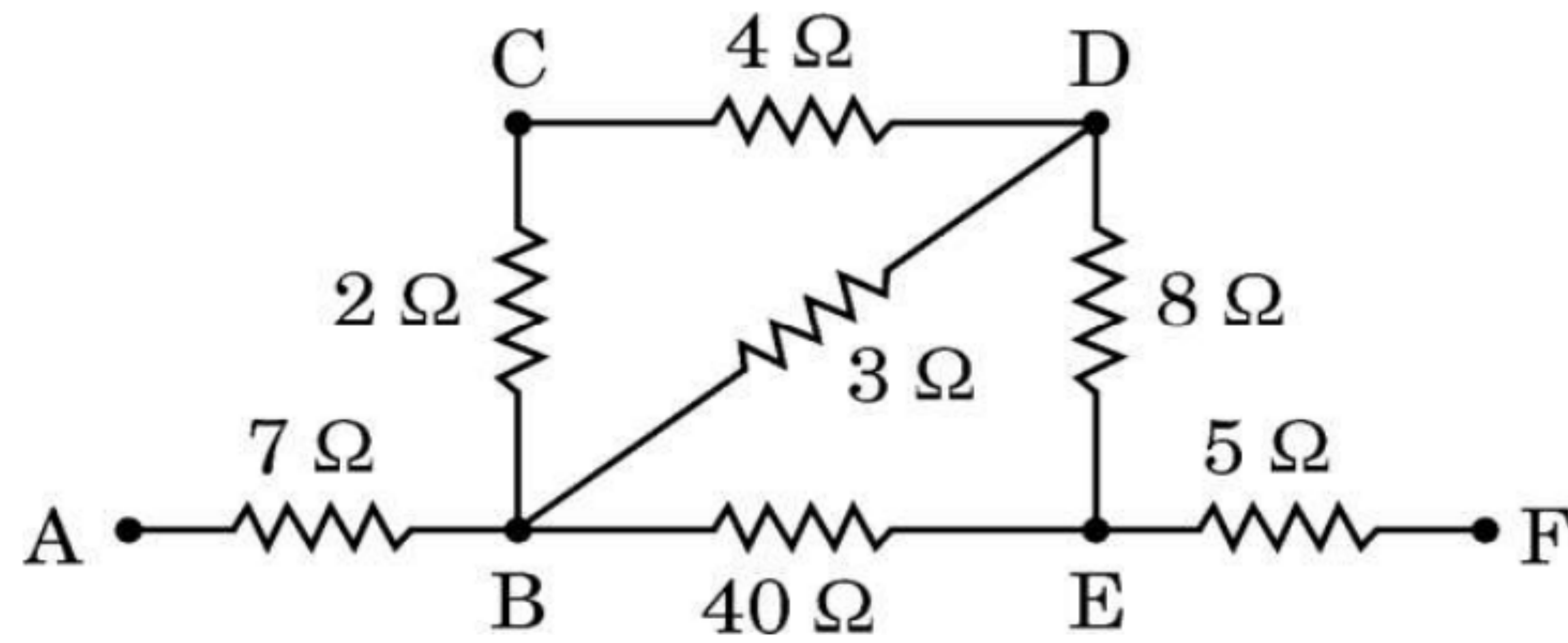
(B) ∞

(C) $\frac{R}{4}$

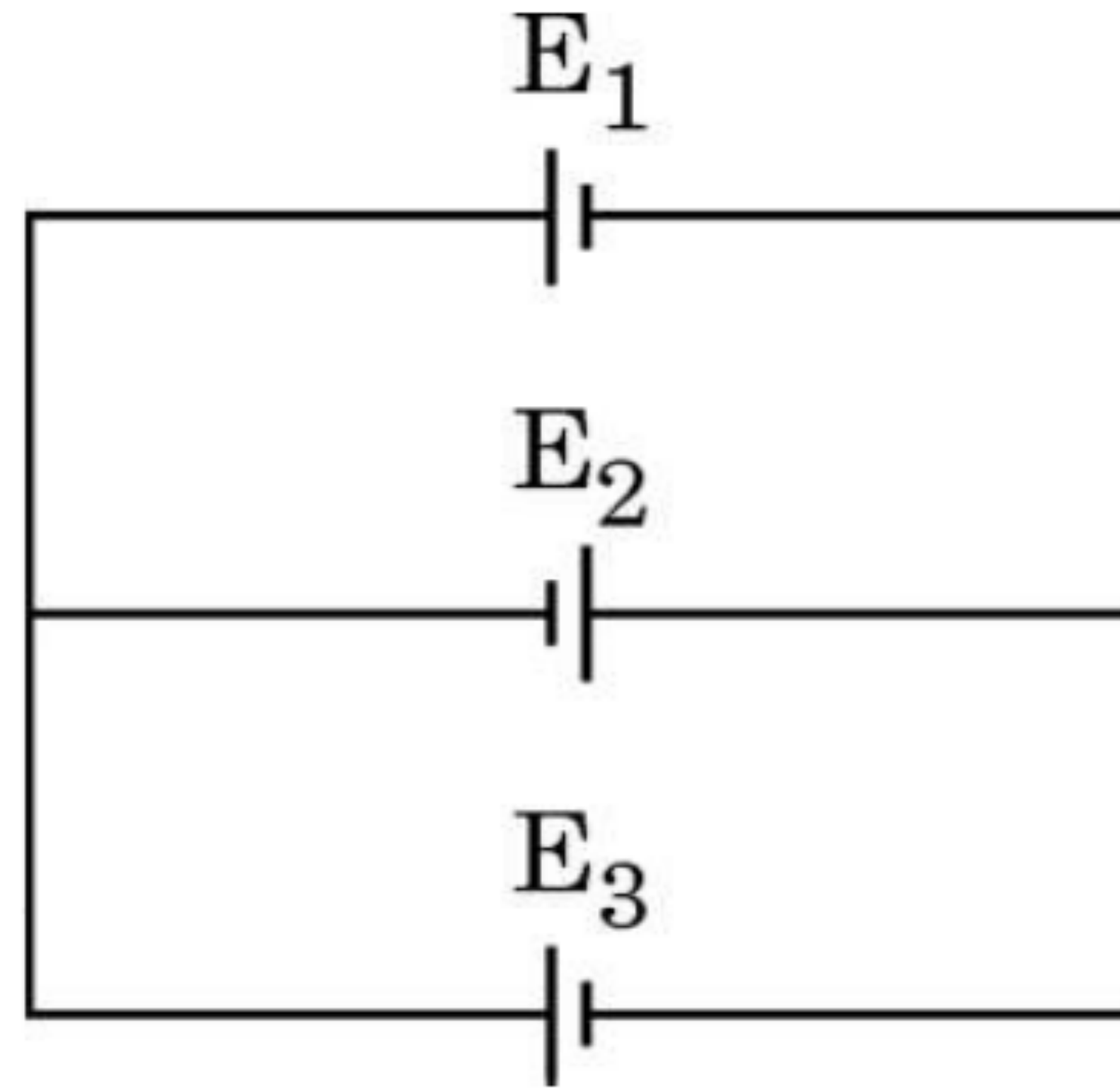
(D) $\frac{4R}{3}$

17. Find the effective resistance of the network of resistors between points A and F as shown in the figure.

2

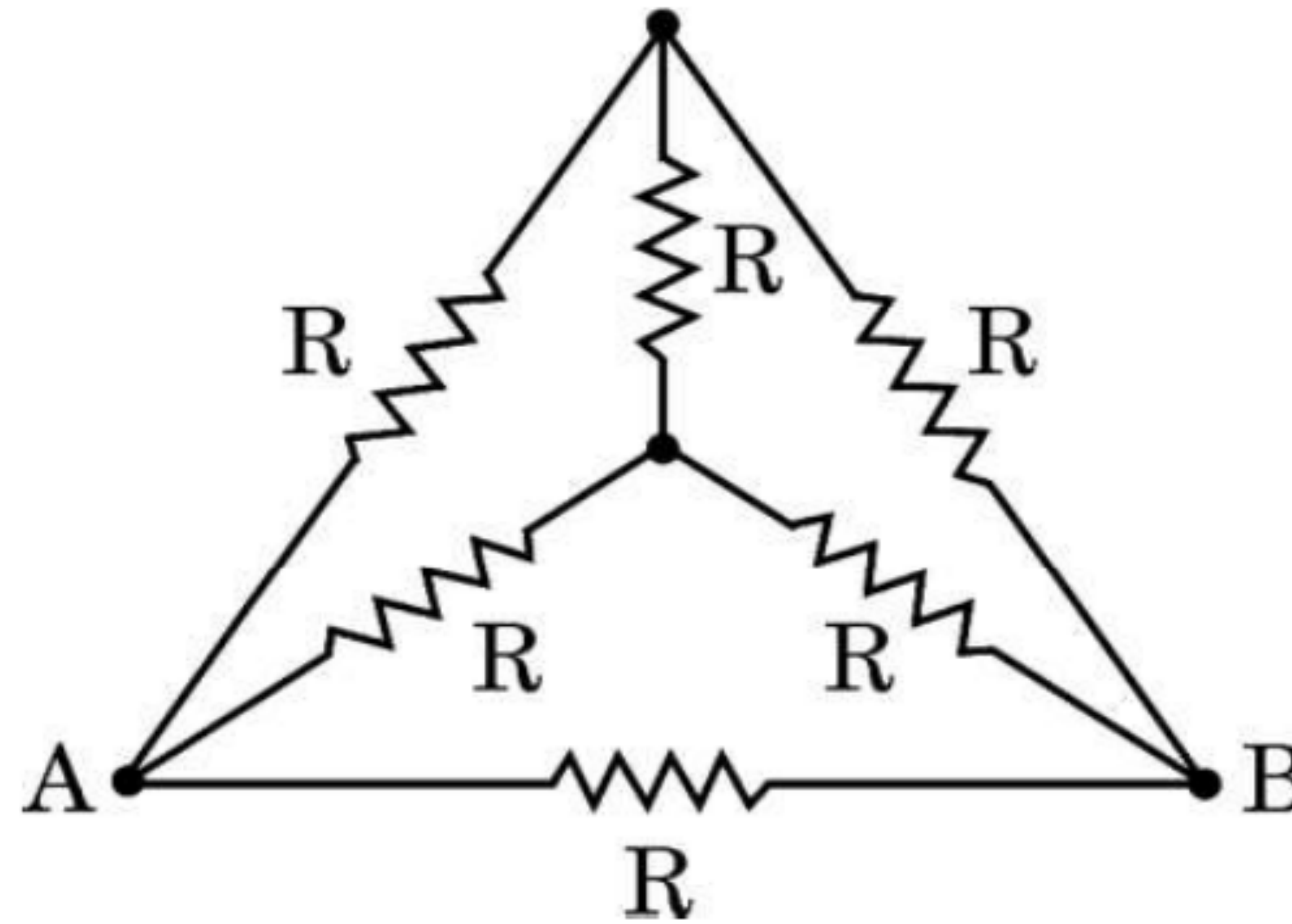


- 32.** (a) (i) Three batteries E_1 , E_2 and E_3 of emfs and internal resistances (4 V, $2\ \Omega$), (2 V, $4\ \Omega$) and (6 V, $2\ \Omega$) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1 , E_2 and E_3 .

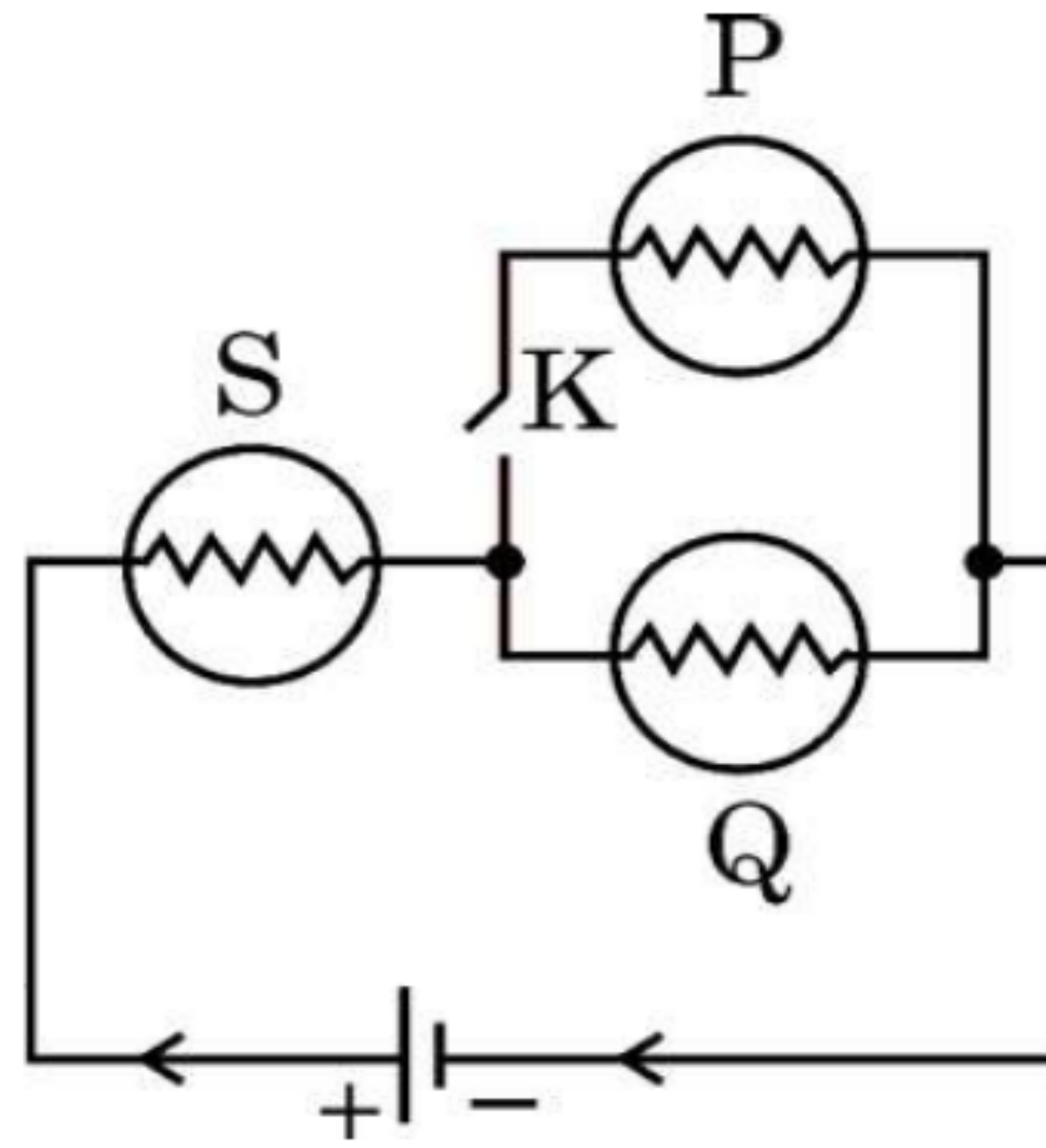


- (ii) The ends of six wires, each of resistance R ($= 10\ \Omega$) are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit.

5



- 21.** (a) In the given figure, three identical bulbs P, Q and S are connected to a battery.

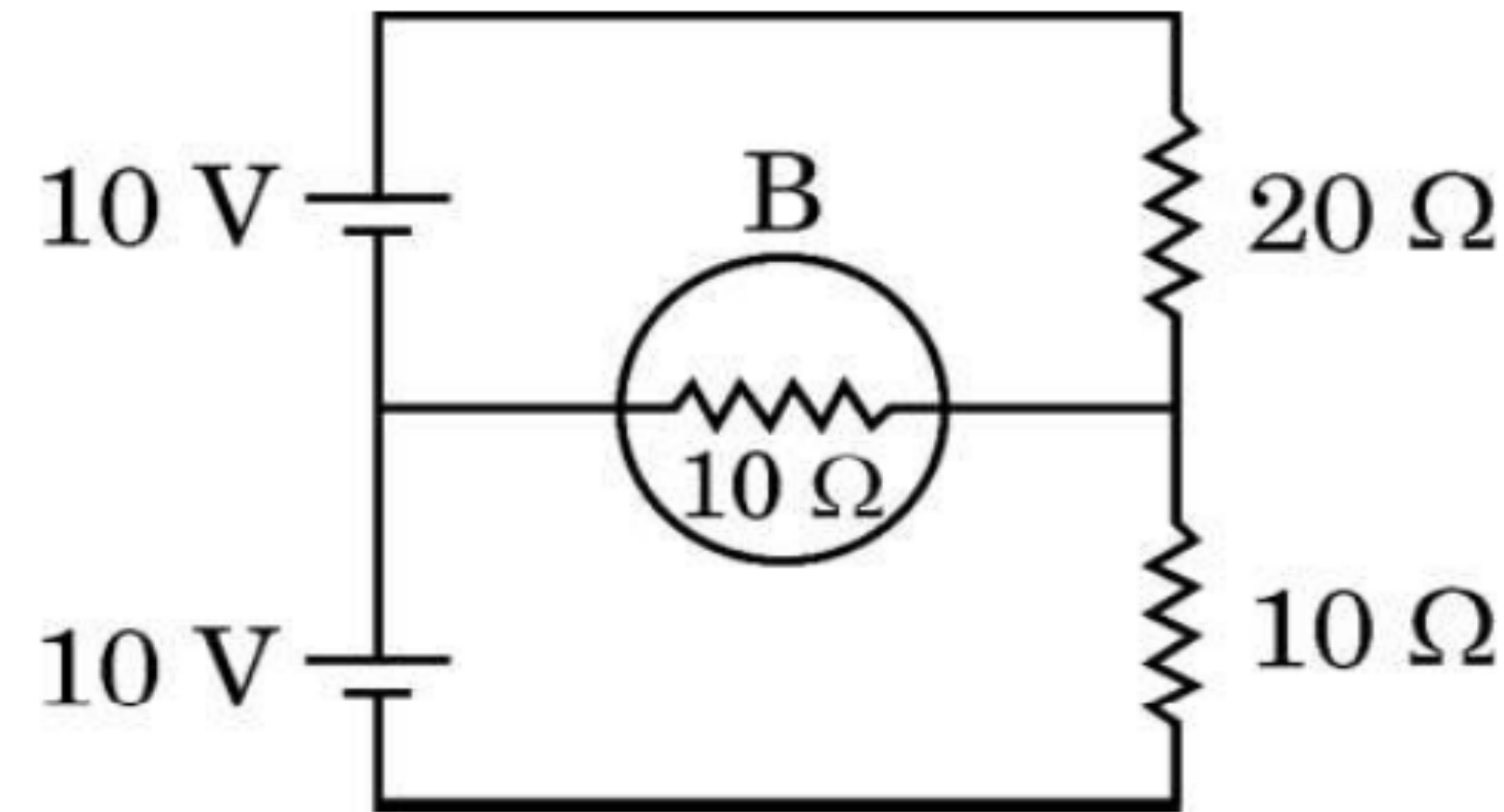


- (i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.
- (ii) Compare the brightness of the bulbs S and Q when the key K is opened.

Justify your answer in both cases.

OR

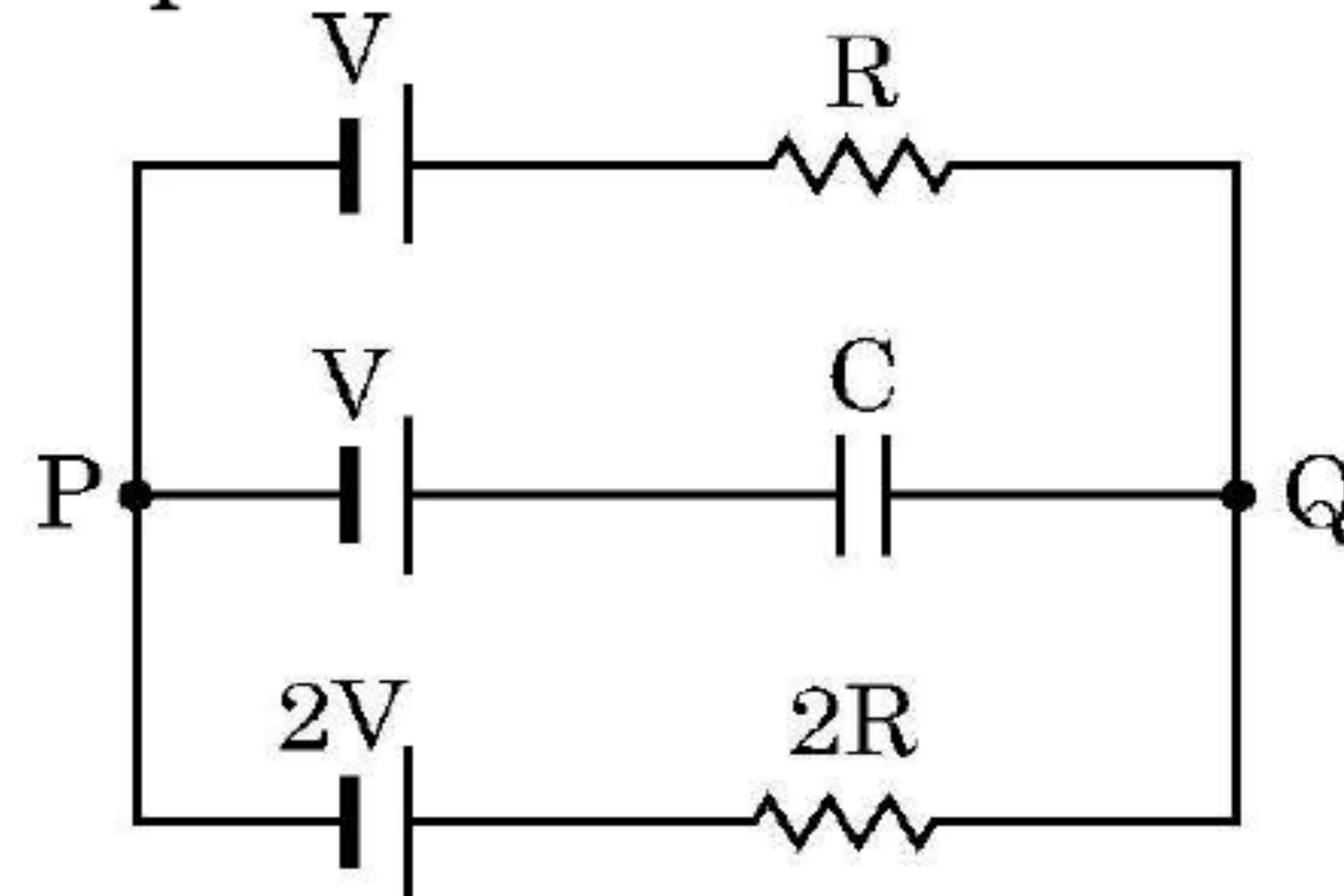
- (b) Two cells of emf 10 V each, two resistors of $20\ \Omega$ and $10\ \Omega$ and a bulb B of $10\ \Omega$ resistance are connected together as shown in the figure. Find the current that flows through the bulb.



21. (a) Two wires of the same material and the same radius have their lengths in the ratio 2 : 3. They are connected in parallel to a battery which supplies a current of 15 A. Find the current through the wires. 2

OR

- (b) In the circuit three ideal cells of e.m.f. V , V and $2V$ are connected to a resistor of resistance R , a capacitor of capacitance C and another resistor of resistance $2R$ as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C .



17. n identical cells, each of e.m.f. E and internal resistance r , are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. **2**

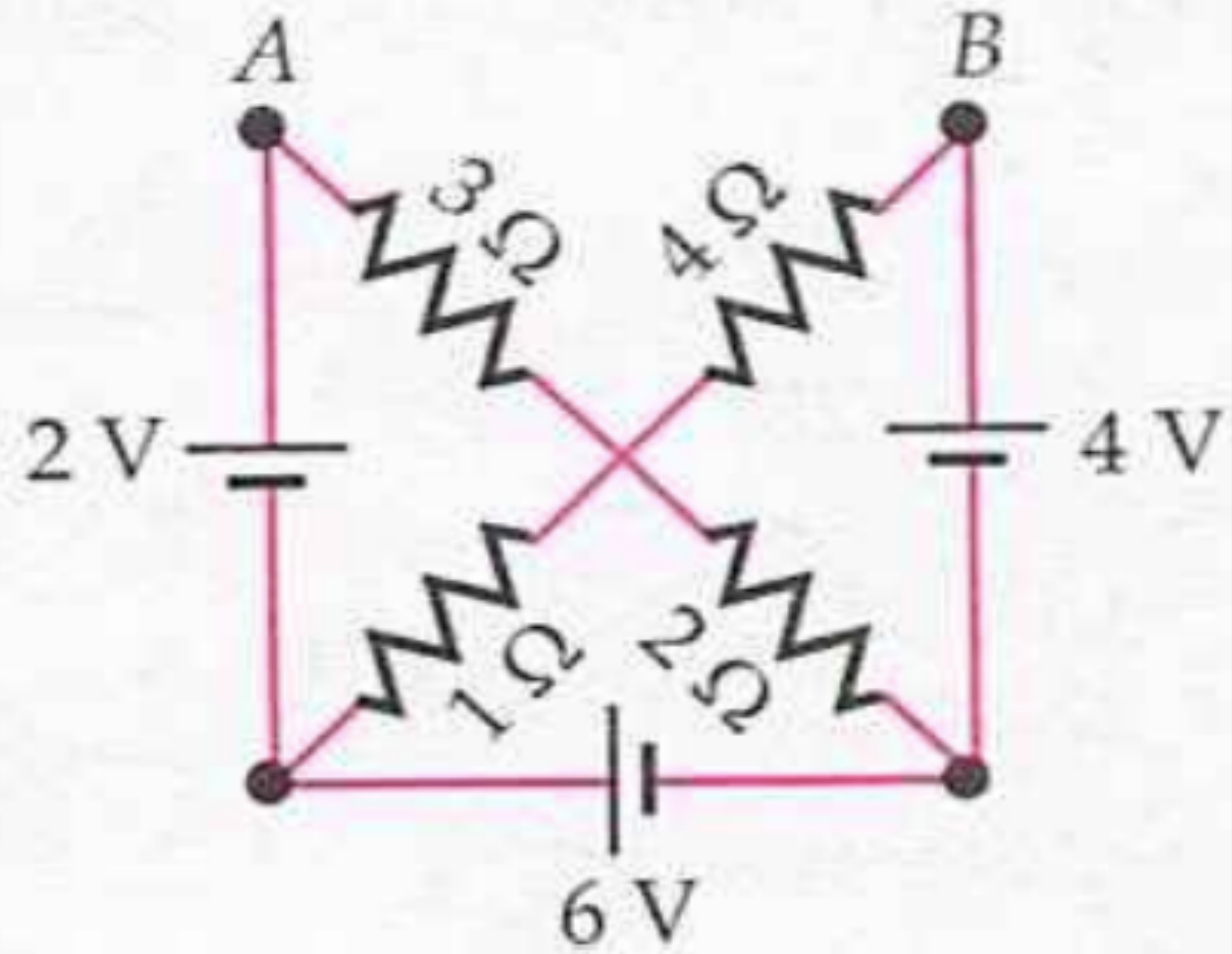
22. (a) Two batteries of emf's 3V & 6V and internal resistances $0.2\ \Omega$ & $0.4\ \Omega$ are connected in parallel. This combination is connected to a $4\ \Omega$ resistor. Find :

3

- (i) the equivalent emf of the combination
- (ii) the equivalent internal resistance of the combination
- (iii) the current drawn from the combination

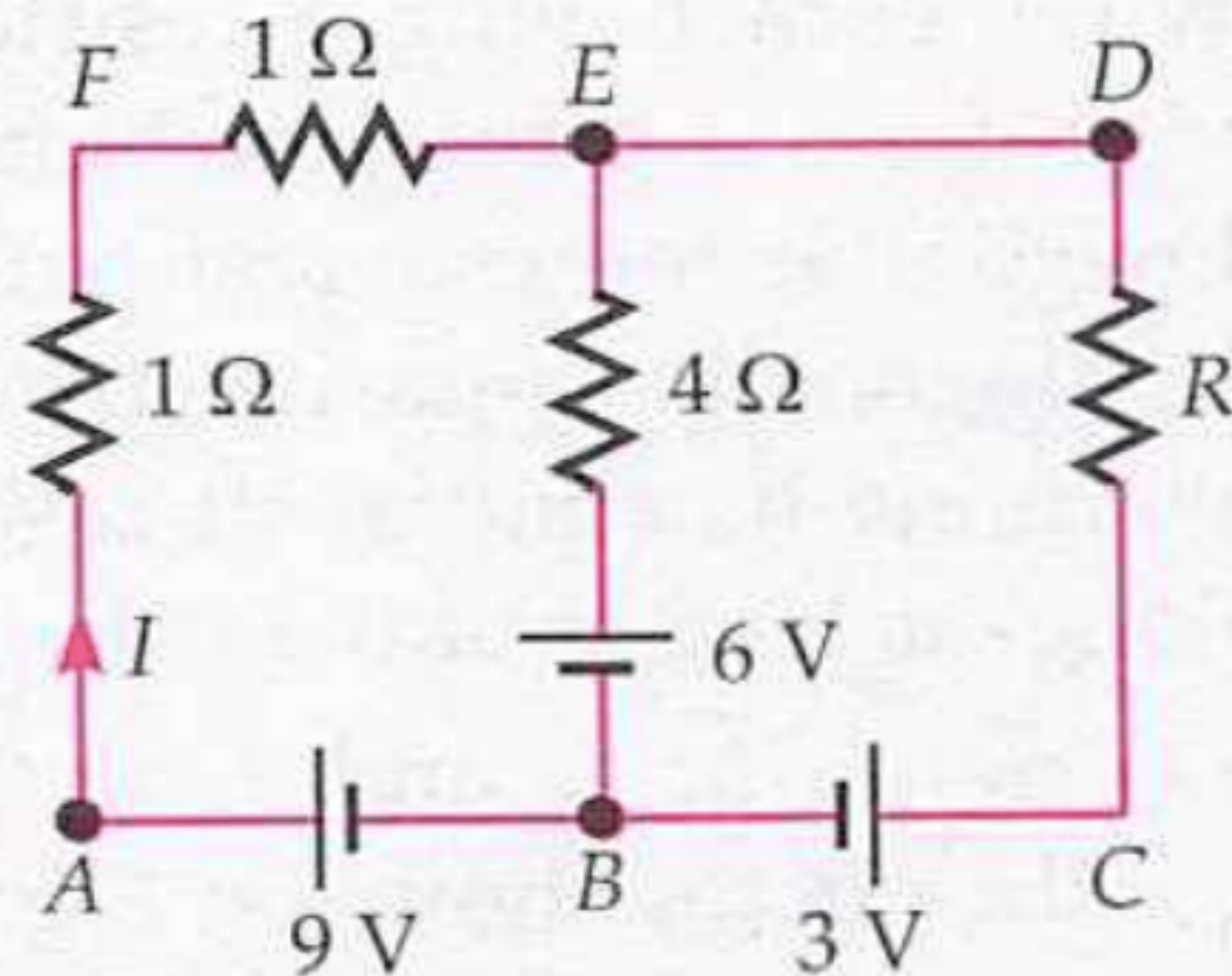
OR

11. In the network shown in Fig. 3.182, (i) calculate the current of the 6 V battery and (ii) determine the potential difference between the points A and B .

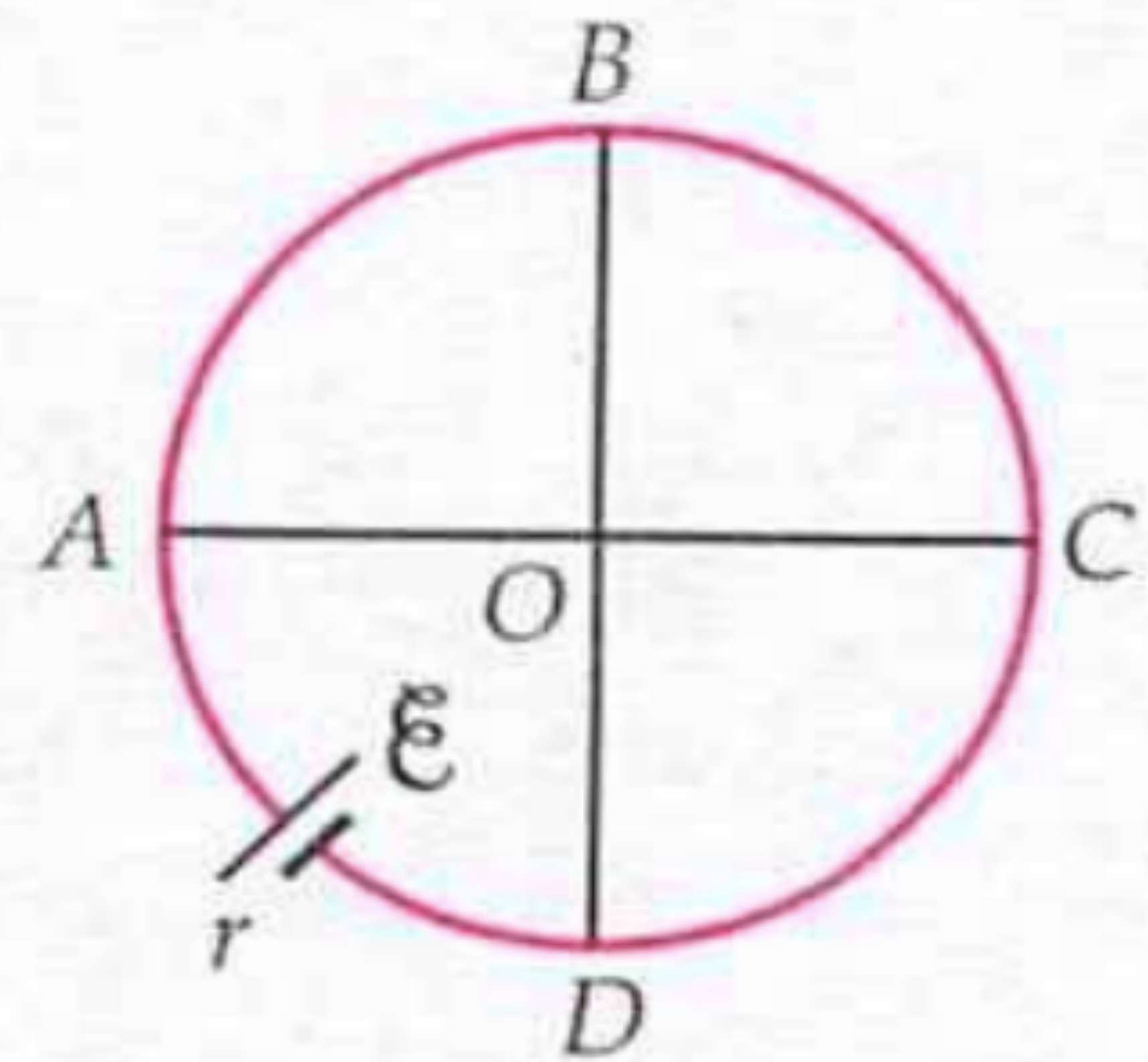


9. Using Kirchhoff's rules, determine the value of unknown resistance R in the circuit shown in Fig. 3.180 so that no current flows through 4Ω resistance. Also find the potential difference between A and D . [CBSE D 12]

(Ans. 3 V)



18. In Fig. 3.188, $ABCD$ is a uniform circular wire of resistance 2Ω . AOC and BOD are two wires along two perpendicular diameters of the circle, each having same resistance 1Ω . A battery of emf $\mathcal{E}$ and internal resistance r is connected between the points A and D . Calculate the equivalent resistance of the network.



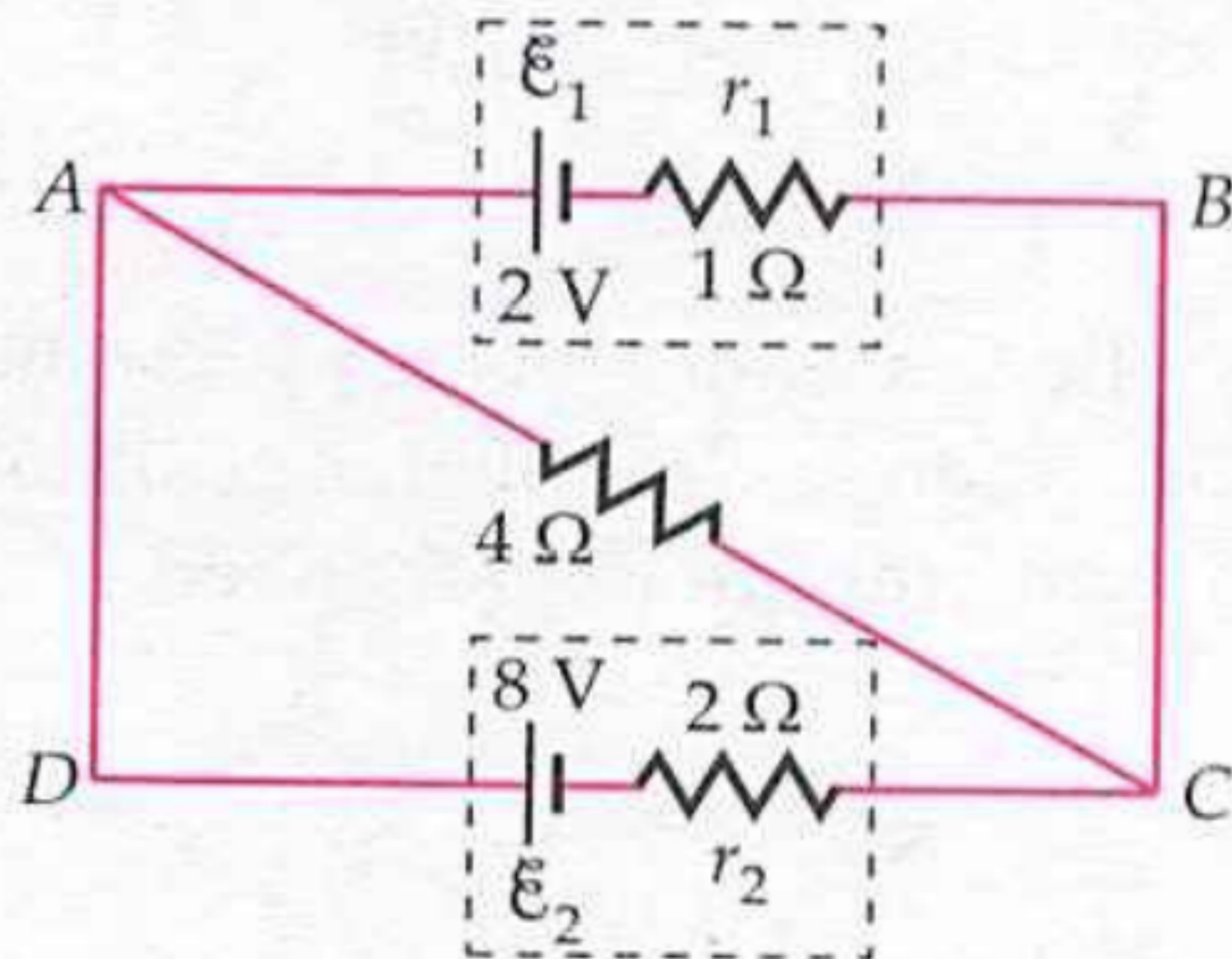
(Ans. $\frac{15}{14}\Omega$)

Fig. 3.188

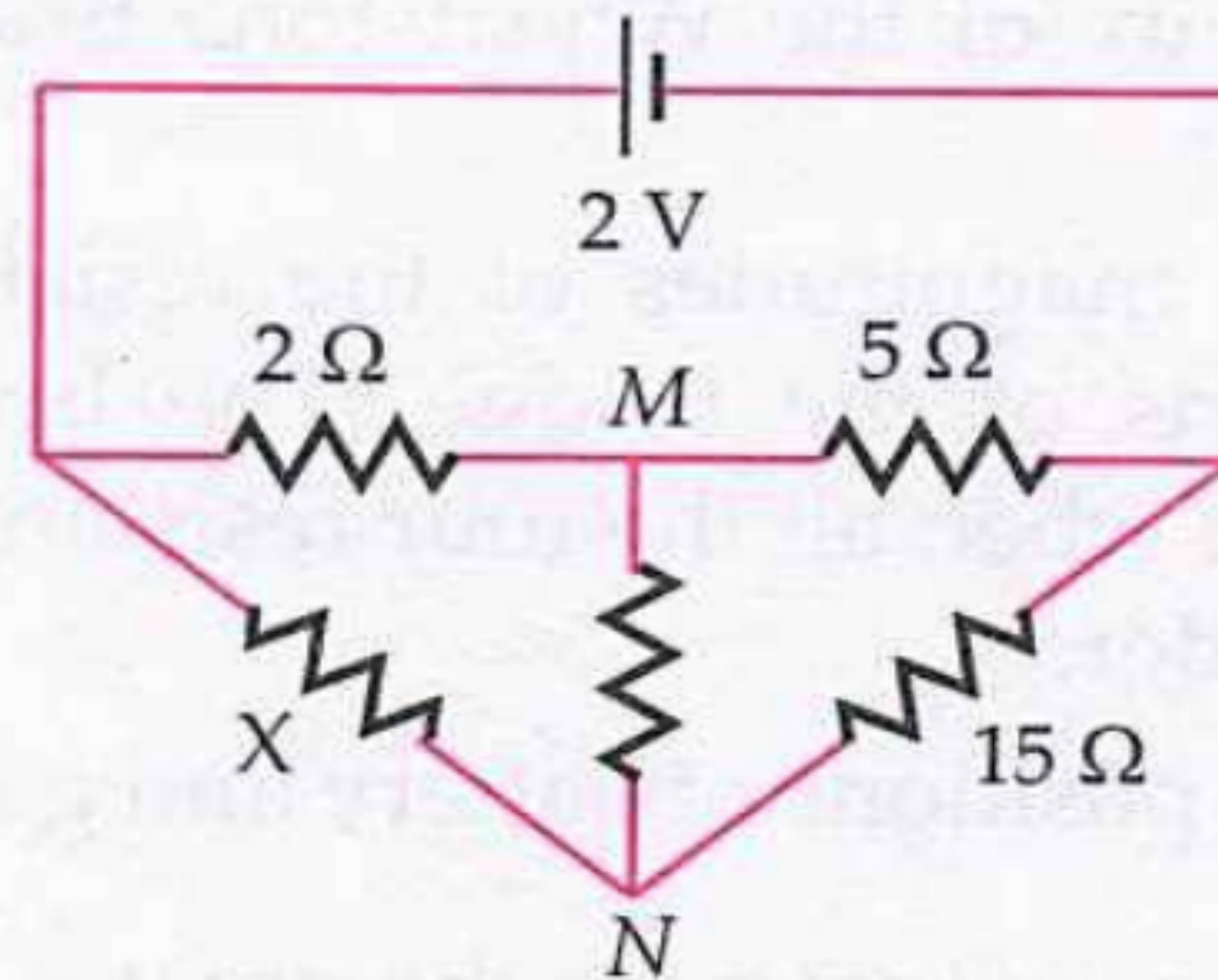
19. A cell of emf $\mathcal{E}_1 = 2 \text{ V}$ and internal resistance $r_1 = 1 \Omega$ is connected to another cell of emf $\mathcal{E}_2 = 8 \text{ V}$ and internal resistance $r_2 = 2 \Omega$ through an external resistance of 4Ω as shown in Fig. 3.189. Find the potential difference between point A and point C.

[CBSE 20C]

(Ans. $\frac{24}{7} \text{ V}$)

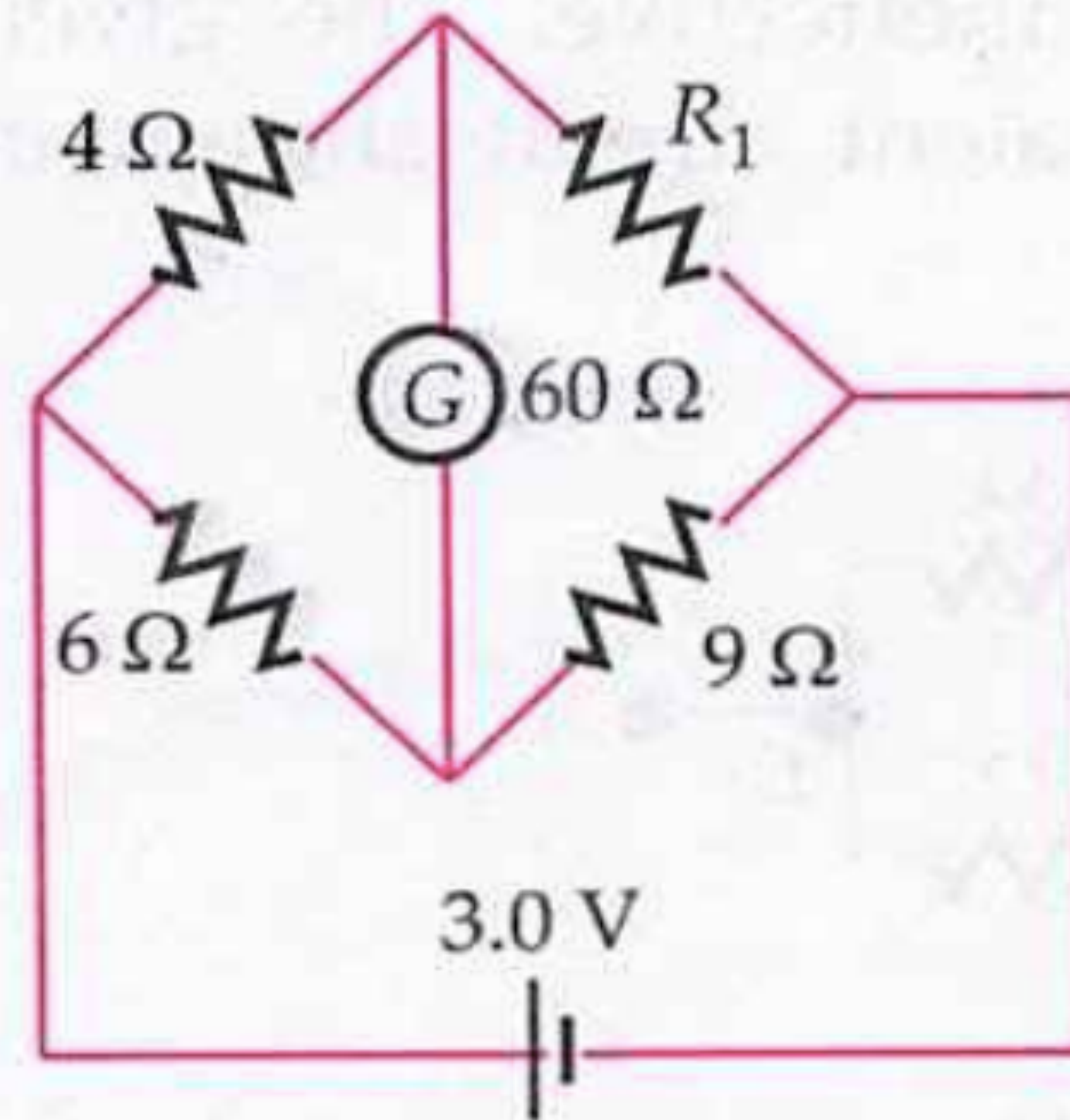


Example 172. Find the value of resistance X in the circuit shown in Fig. 3.218 so that the junctions M and N are at the same potential. [ISCE 09]

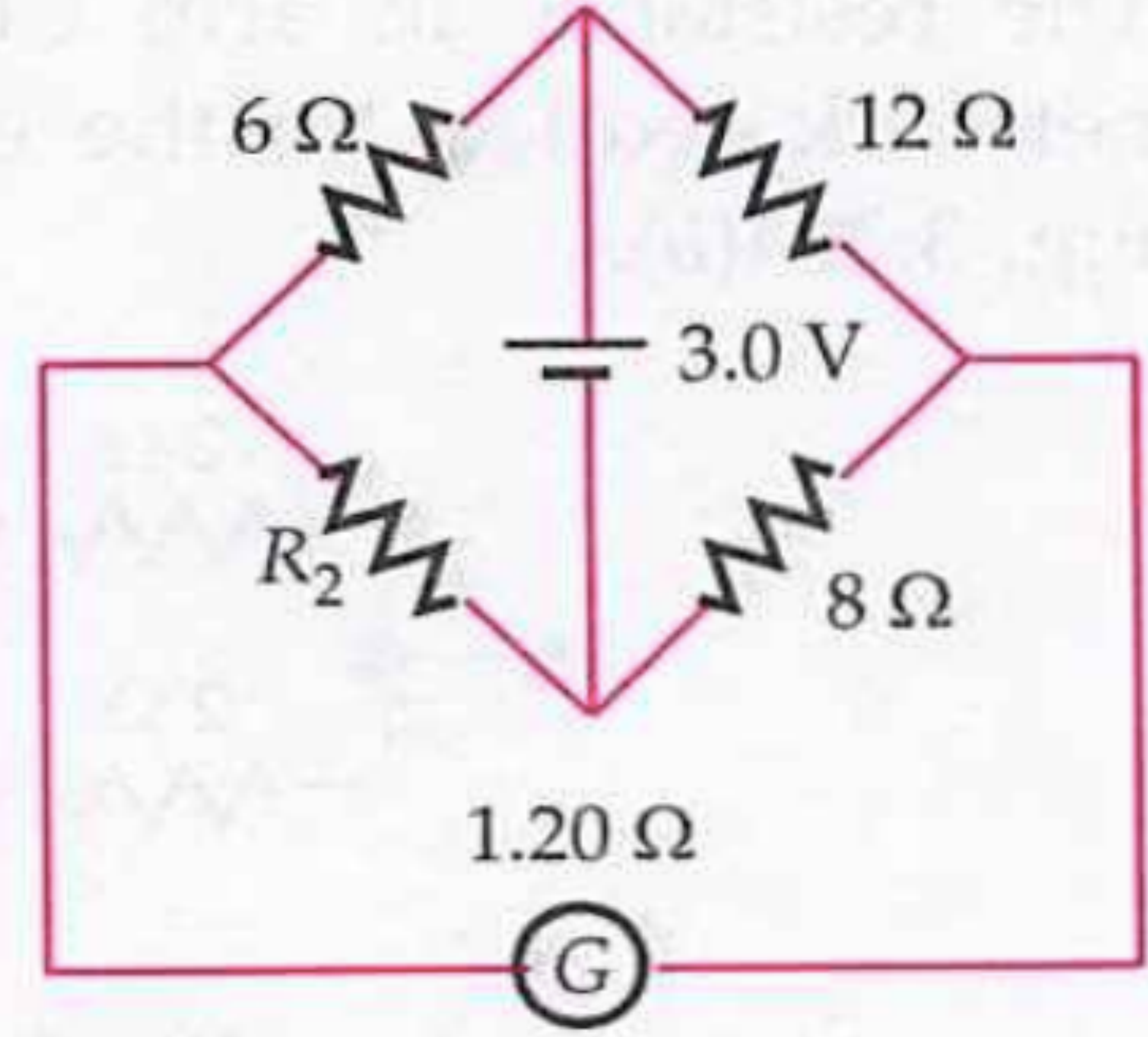


Example 175. The galvanometer, in each of the two given circuits, does not show any deflection. Find the ratio of the resistors R_1 and R_2 , used in these two circuits.

[CBSE OD 13]



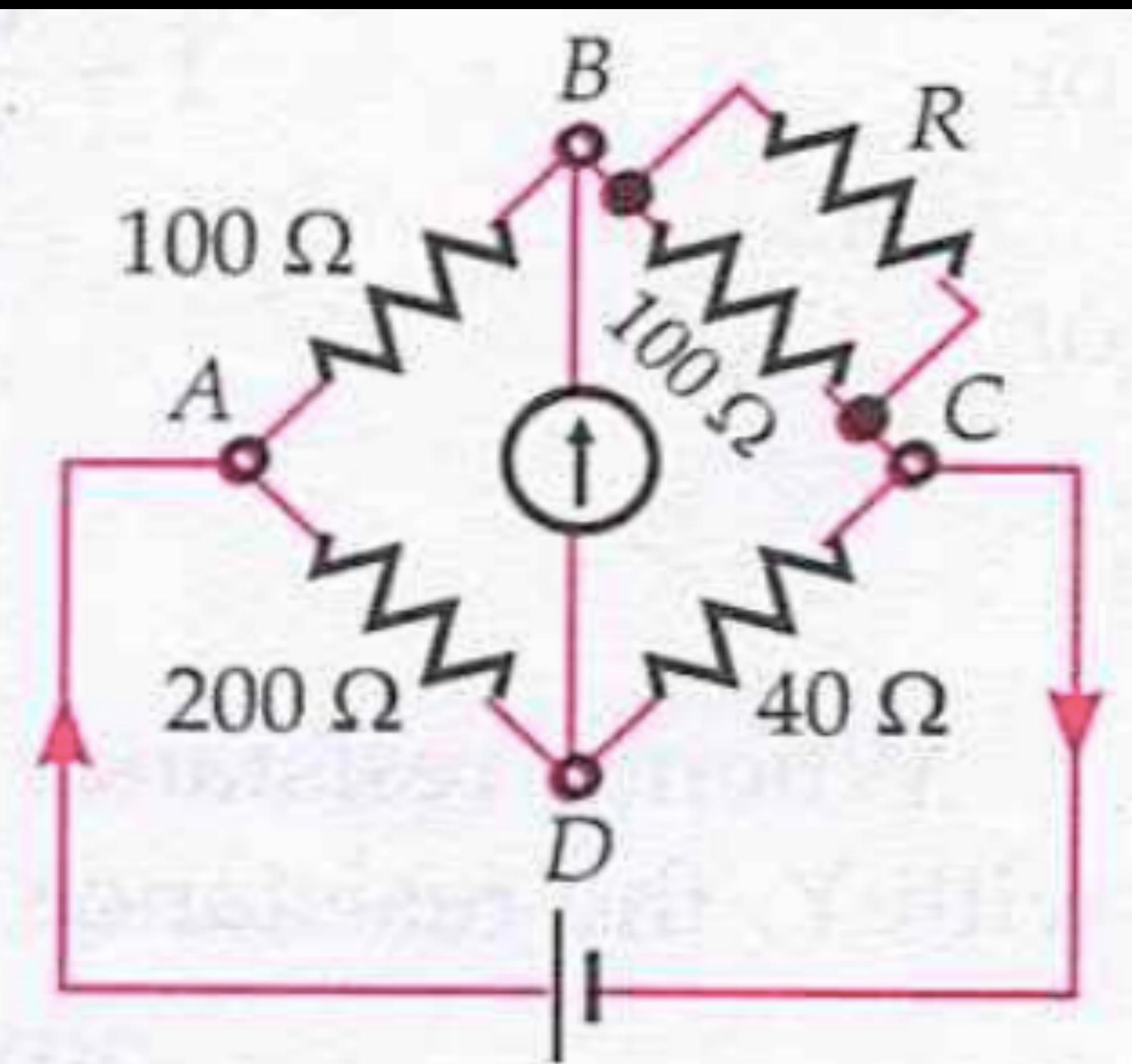
Circuit 1



Circuit 2

1. The Wheatstone's bridge of Fig. 3.234 is showing no deflection in the galvanometer joined between the points B and D . Compute the value of R .

(Ans. $25\ \Omega$)



12. Find the equivalent resistance between the points X and Y of the network shown in Fig. 3.245.

